Audio/Home TechOps | SW DFM ***v2.8***

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RFP Rev	Change Description	Release Date	Updated by
v1.0	Initial release	Nov 4. 2022	Chengshi Miao
v1.1	Add HIVE requirements	Dec 8. 2022	Chengshi Miao
v1.2	Add color code definition. Add another tab under HOME for semi-auto stations. Show top10 in pie chart on ALARM tab. Show SN in I/O card. Add definitions on data format. Add rules on sensor disablement. And other minor changes	Dec 14. 2022	Chengshi Miao
v2.0	Soft reset. Machine log format definition. Remote qualification definition. HIVE data reporting rule. Setting page definition. And other minor changes	Feb 13. 2023	Chengshi Miao, Michael Northrop
v2.1	Add requirements on stations using HIVE to upload pdata and blob data to Insight.	Feb 25. 2023	Chengshi Miao
v2.2	Add station card background requirements under remote qualification mode. Add requirements to record SN and operator ID for all manually triggered planned downtime. Add specific vision page for cosmetic AOI.	Mar 13. 2023	Chengshi Miao
v2.3	Move remote qualification folder to HIVE log folder. Change login page requirement, remove passwords. All manual action except for pressing E-Stop and switch state from running to idle will need to read card UID and upload to HIVE. GUI enable home, alarm, data and vision when no login, but still in read only. Add access control requirements. Update vision page for cosmetic AOI. Change the size of HIVE status bar for engineering, planned downtime and downtime. Update HIVE state switch definition.	Apr 4. 2023	Chengshi Miao
v2.3.1	Add GUI definition on how to log out current account	Apr 6. 2023	Chengshi Miao
v2.4	Add support for uploading test head ID and fixture ID with HIVE. Add requirements to collect tossing data. Add new planned downtime error message for glue weighing. Add new key to record dispenser purge frequency in HIVE machine data. Add requirements to display all Insight data in Home->Parameter Dashboard. Add software requirements to prevent manual key-in of ID and SN.	May 6. 2023	Chengshi Miao, Nick LaBarbera
v2.5	Add LAD to planned downtime. Optimize popup window for downtime. Change the key name “badge” in HIVE machine state to “u_id”.	July 20, 2023	Chengshi Miao
v2.6	Relayout the sides. Add unit_ct data collection	Sep 22, 2023	Chengshi Miao, Kyle Hu
v2.7	S1.1 Add labels in main UI to show current config (show station ID for common products) S1.2 Add key SoftwareType to distinguish between main and vision software S1.3 Add planned downtime codes PD-12, PD-13, PD-14 S1.3.4 Add a tab for the tossing dashboard. Enable switching between daily and hourly data viewers. Add a summary for tossing data S1.3.6 Add product type on the settings page S1.4 Update color code standards for newly added icons S2.1 Add explanations for new planned downtime codes S2.7 Set CT upper limits to 3600. The initial value is set to 0 instead of -1 Add a table of contents Add S3.1, S3.3, S3.4, S3.5 for explanations on PD-04 (valve change), PD-12 (nozzle change), PD-13 (CCD Repeatability Testing), and PD-14 (Dispensing Pose Calibration)	Dec 26, 2023	Chengshi Miao
v2.7.1	S2.2 Add to collect Mac address for better support remote qualification	Mar 28, 2024	Chengshi Miao

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RFP Rev	Change Description	Release Date	Updated by
v2.8	S1.1 Update descriptions on machine log content format Add S1.2.1 on 2 step verification for SI Level 3 access S1.3.2 Update to add a pop window to collect card id and refresh machine state payload to HIVE. This is an intend to collect OSS response time S2.1 Following updates in S1.3.2, update state change rule from state 5 accordingly	May 22. 2024	Chengshi Miao

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S3.5	Dispensing Pose Calibration (PD-14)	Only for SCUD

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S1-Machie GUI

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Home Buttons

	Energized	Alarmed	De-energized	Disabled	
Home					Always be enabled Read only
Alarm					Always be enabled Turn to Alarmed when an error occurs
Config					No login: **Disabled** L1, 2, 3: Enabled
Data					Always be enabled Read only
Vision					Always be enabled Read only
Setting					No login: **Disabled** L1, 2, 3: Enabled
Running					No login: Enabled L1, 2, 3: Enabled
Paused					No login: Enabled L1, 2, 3: Enabled
Stopped					No login: Enabled L1, 2, 3: Enabled

User Level

No login		
L1		User ID is logged. Normal production access. All parameters are read-only and cannot be changed
L2		Engineer ID is logged. Universal change access. Key parameters cannot be changed
L3		Admin ID is logged. Can change all parameters. All actions must be recorded by log

**For Level 2 and Level 3 users, if no operation is performed within 5 minutes, the system automatically logs out the user*

**Level 3 user must be approved by Apple DRI before granting access*

HIVE Status

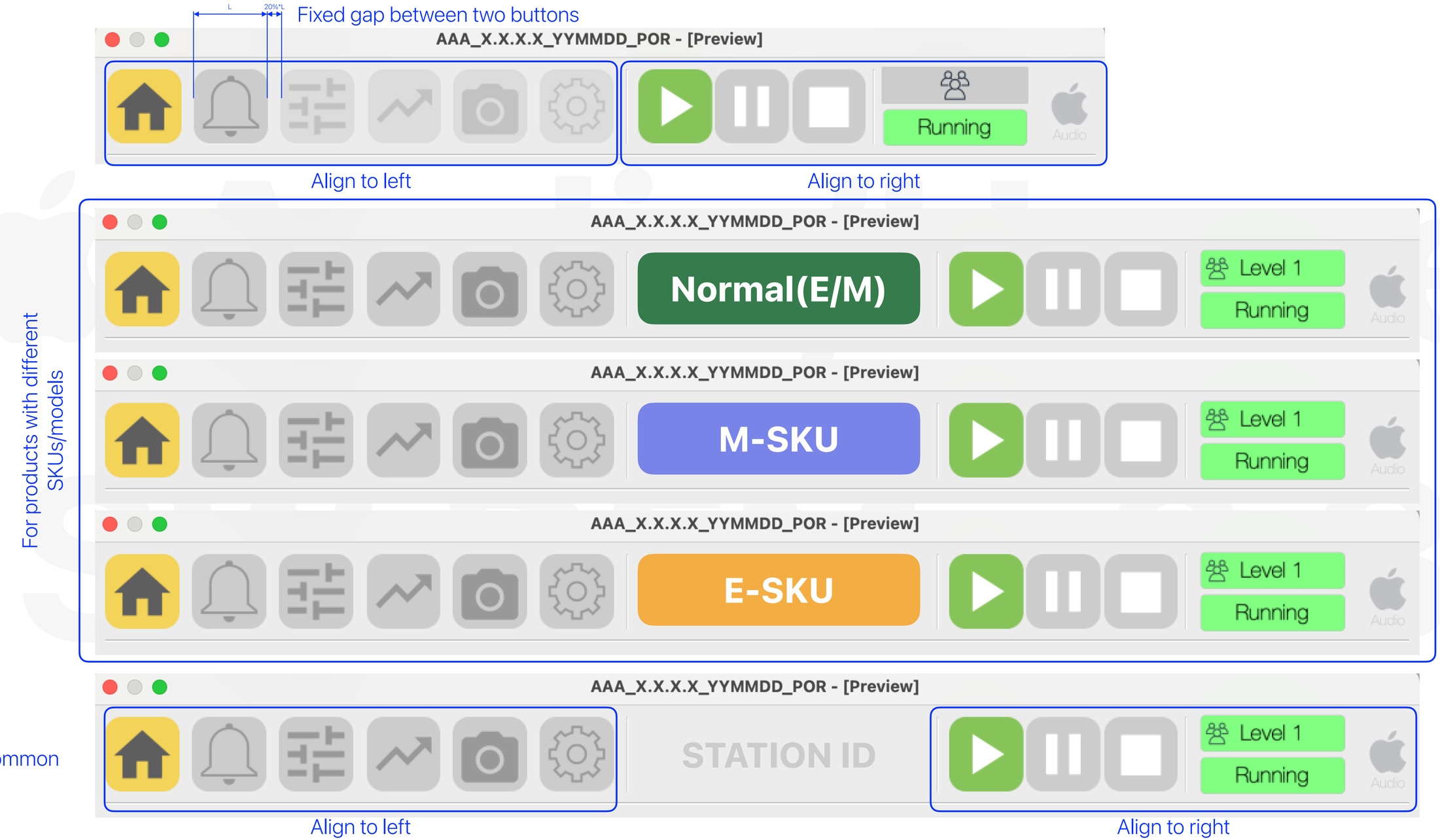
	HIVE is disabled or disconnected
	HIVE state 1: Running
	HIVE state 2: Idle
	HIVE state 3: Engineering
	HIVE state 4: Planned downtime
	HIVE state 5: Downtime

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S1.1-No login



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Home, alert, data and vision buttons are enabled. The rest of the buttons are **disabled** before login

Machine running/pause/stop buttons are enabled

Hive status bar shows current status. View only



Machine is running. Long press to reset(need popup window for confirmation)



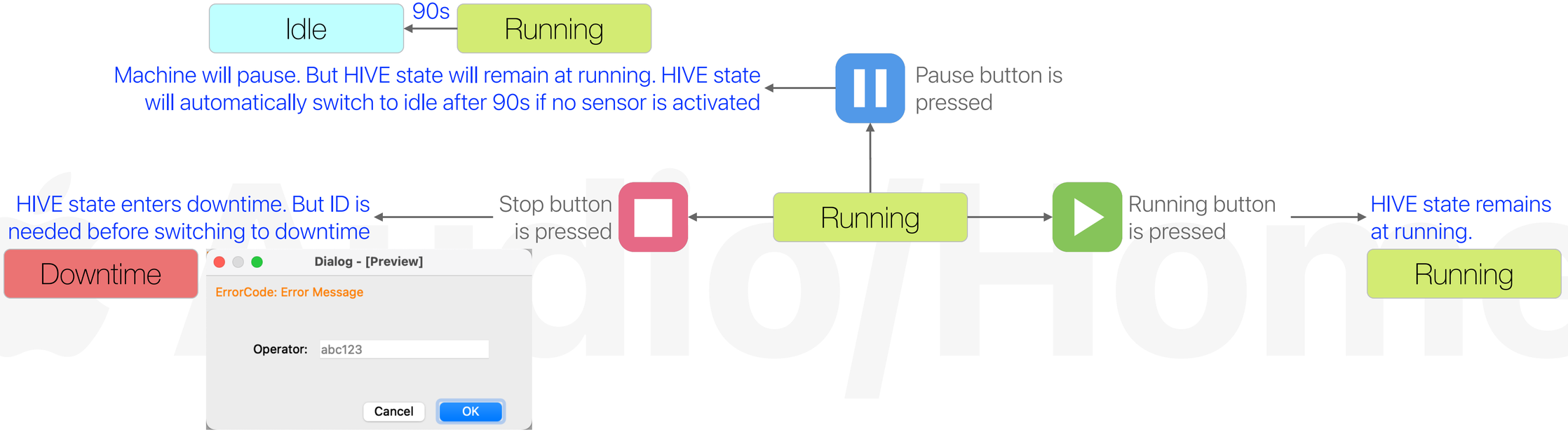
Machine paused

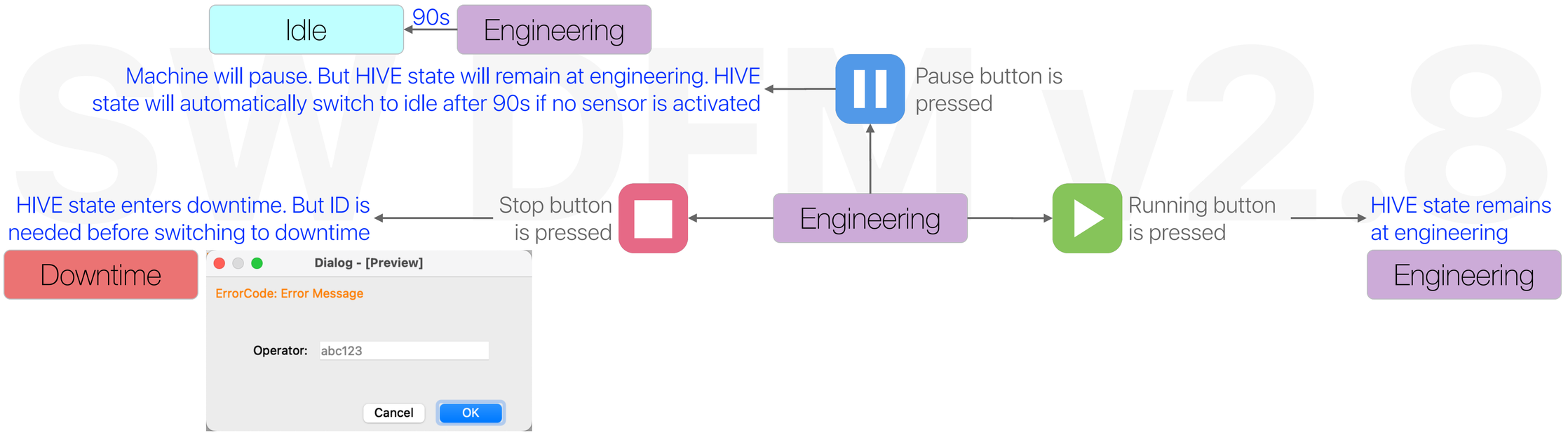


Machine stopped

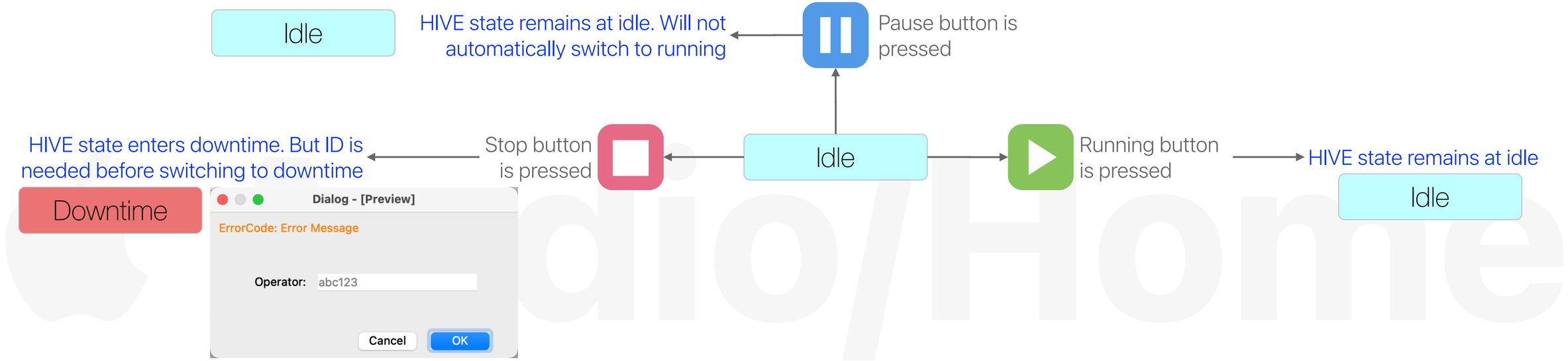
HIVE	HIVE is disabled
Running	HIVE state 1: Running
Idle	HIVE state 2: Idle
Engineering	HIVE state 3: Engineering
Planned DT	HIVE state 4: Planned downtime
Downtime	HIVE state 5: Downtime

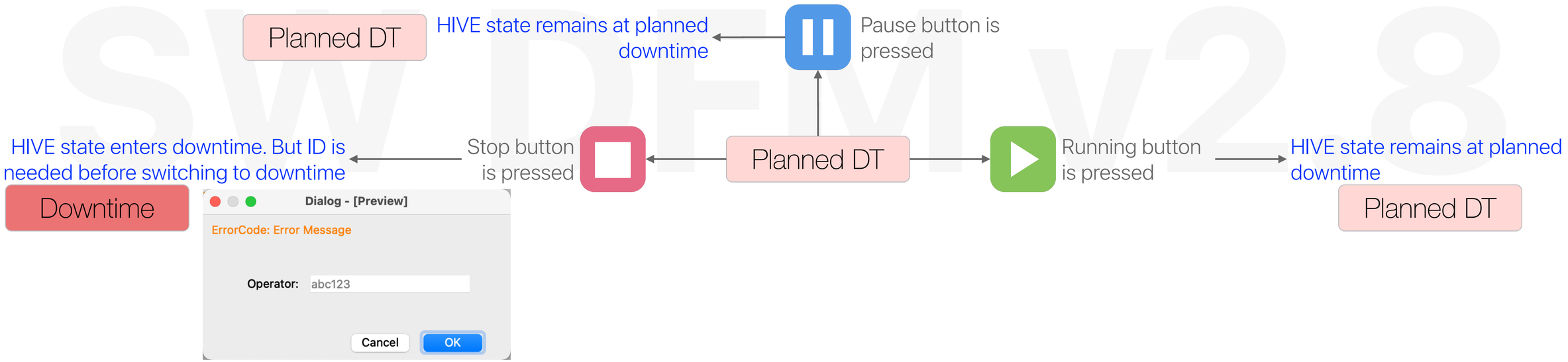
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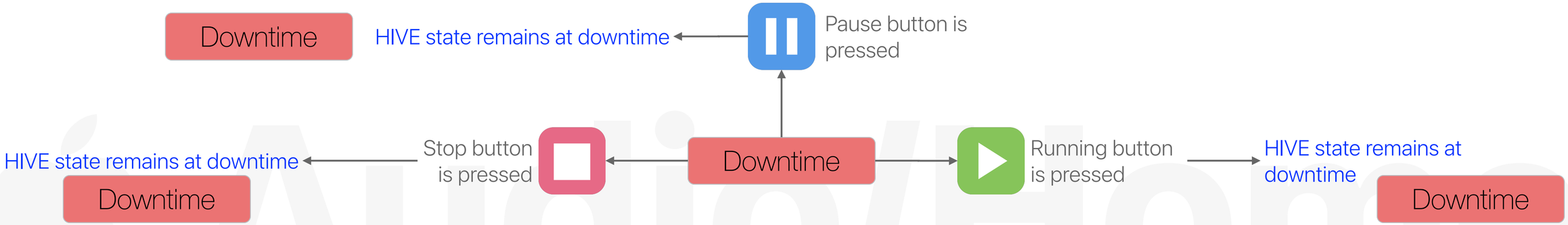


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1. Stop button work the same as physical E-STOP button. But every time when pressing the stop button on GUI will pop up to record operator ID. Pressing physical E-STOP will stop the machine and change HIVE state immediately
2. At HIVE running, idle, engineering, planned downtime or downtime state, pressing running or idle button will not change HIVE state

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HIVE state: **running**



HIVE state: **idle**



HIVE state: **engineering**



HIVE state: **planned downtime**



HIVE state: **downtime**

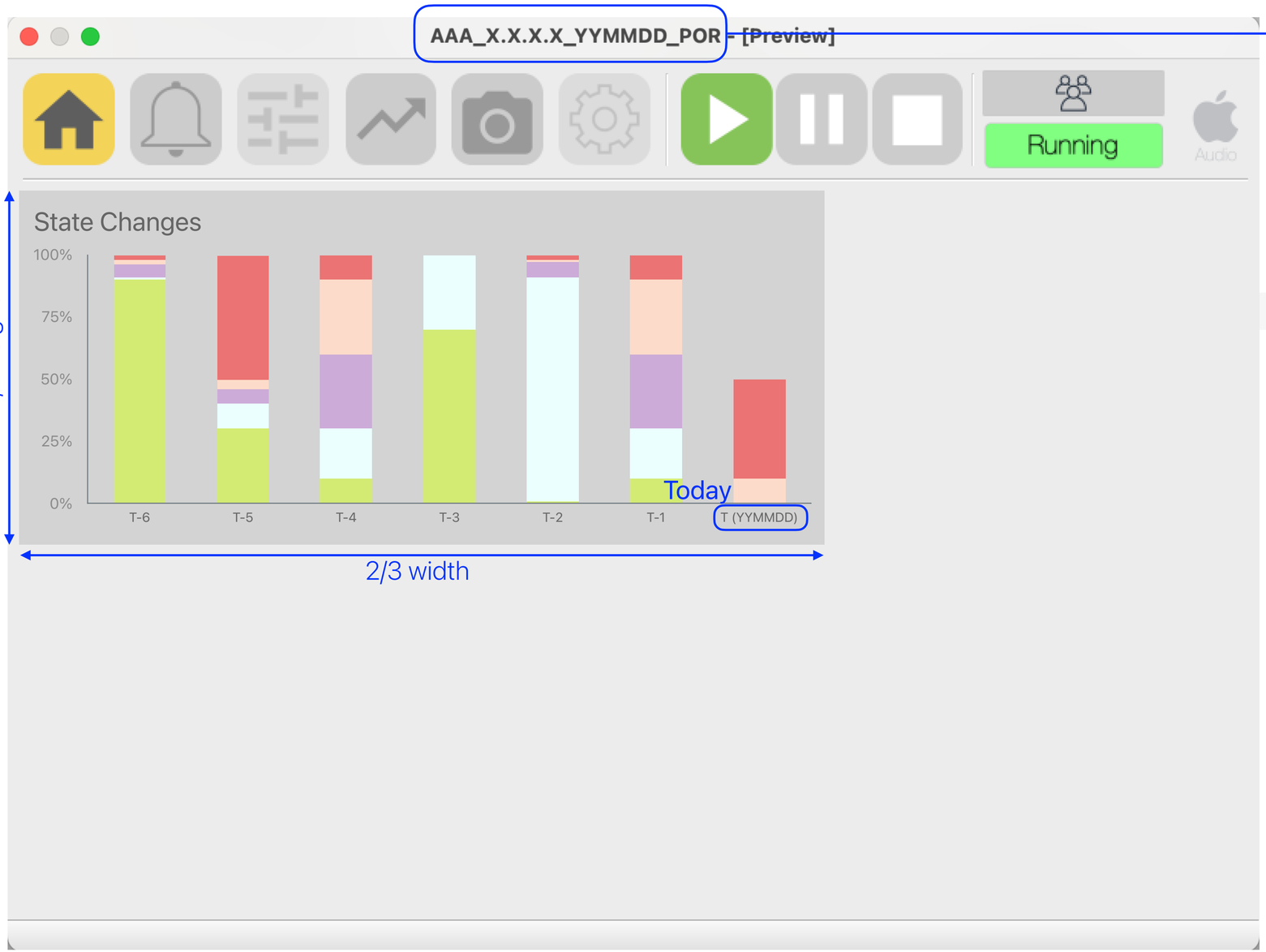
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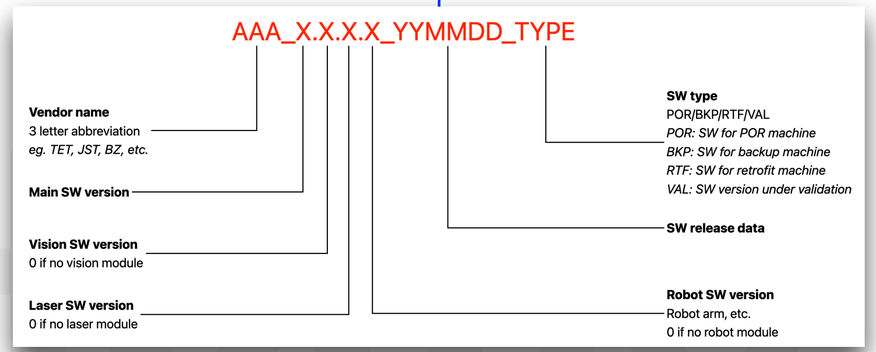
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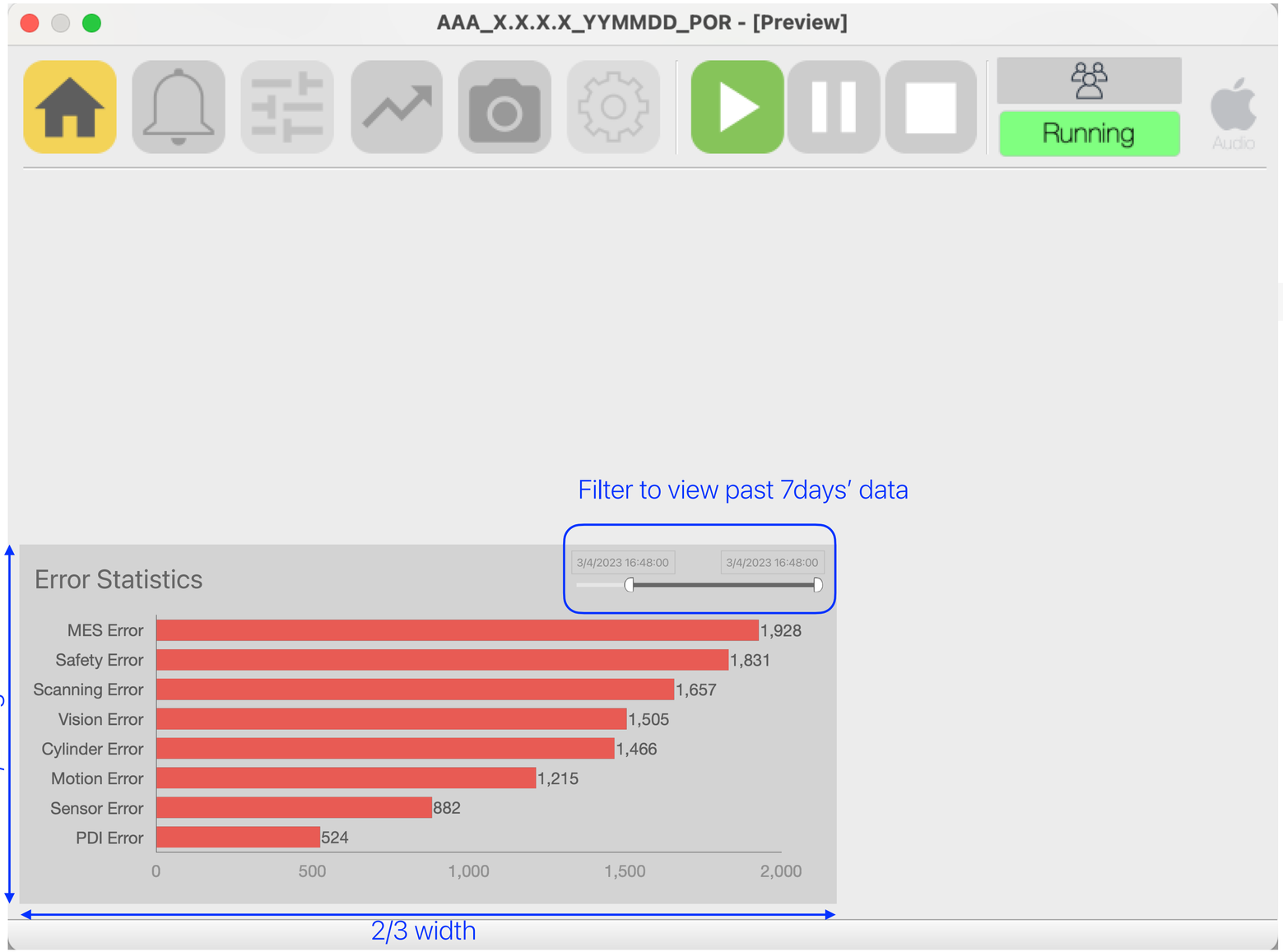




Key Requirements

- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** *when no login*
- State Change dashboard shows past 7 days' state data

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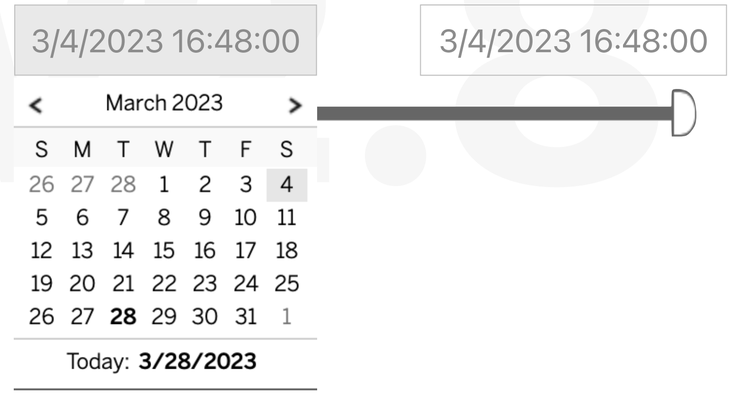


Key Requirements

- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** *when no login*

- Error Statistics dashboard show today's data, store past 7days' data locally
- Can filter to view past 7 day's data





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Key Requirements

- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** when no login
- Show last unit if machine is idle

- I/O Summary updates date once one units if processed
- I/O Summary data shows as hourly

FVMDJ20X6G — Unit FVMDJ20X6G passed

FVMDJ20X6G — Unit FVMDJ20X6G failed

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Key Requirements

- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** when no login

- Put OOS params on top
- Follow AME definition on critical params
- Use N/A in LSL/USL if no spec limitation
- Show current glue SN with no limits (dispenser only)
- Show all pdata uploaded to Insight

#	Value of of spec
#	Value in spec

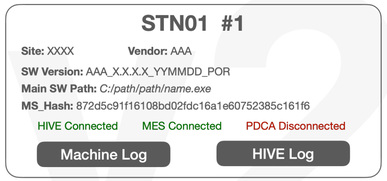
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Key Requirements

- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** when no login

- Station card to show station name, QPL #, Site, Vendor information
- Version code hard coded in codes and shows on station card
- Main SW path to show the path of the main SW, click to open the path
- Update MS Hash key once machine state changes
- Click Machine log and HIVE log to open the path of the log files



- ***Normal***



- ***Remote qualification*** *(disable by default in normal production, unless be notified/ requested to enable by Apple)*

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HIVE Log

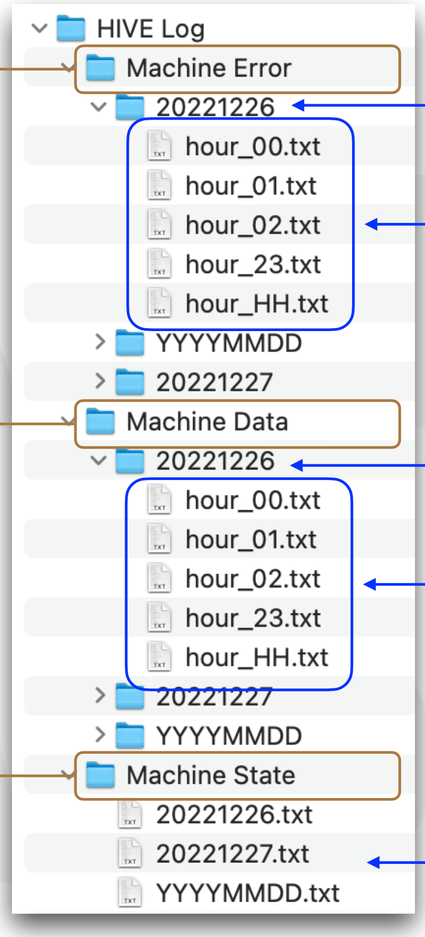
- HIVE Log
  - Machine Error
    - 20221226
      - hour_00.txt
      - hour_01.txt
      - hour_02.txt
      - hour_23.txt
      - hour_HH.txt
    - YYYYMMDD
    - 20221227
  - Machine Data
    - 20221226
      - hour_00.txt
      - hour_01.txt
      - hour_02.txt
      - hour_23.txt
      - hour_HH.txt
    - 20221227
    - YYYYMMDD
  - Machine State
    - 20221226.txt
    - 20221227.txt
    - YYYYMMDD.txt

Machine Log

- Machine Log
  - 20221227
    - hour_00.txt
    - hour_01.txt
    - hour_02.txt
    - hour_23.txt
    - hour_HH.txt
  - 20221226
  - YYYYMMDD

HIVE Log: Record each json file contents sent to HIVE and message received from HIVE

HIVE log includes **Machine State**, **Machine Error** and **Machine Data**. Each stored in separated folder



Machine error sorts by date folders. Naming rule: **YYYYMMDD**. For example 20221227

Each hourly log is recorded in one log file. Naming rule: **hour_HH**. For example hour_00, meaning this file is the machine log from 00:00 to 01:00

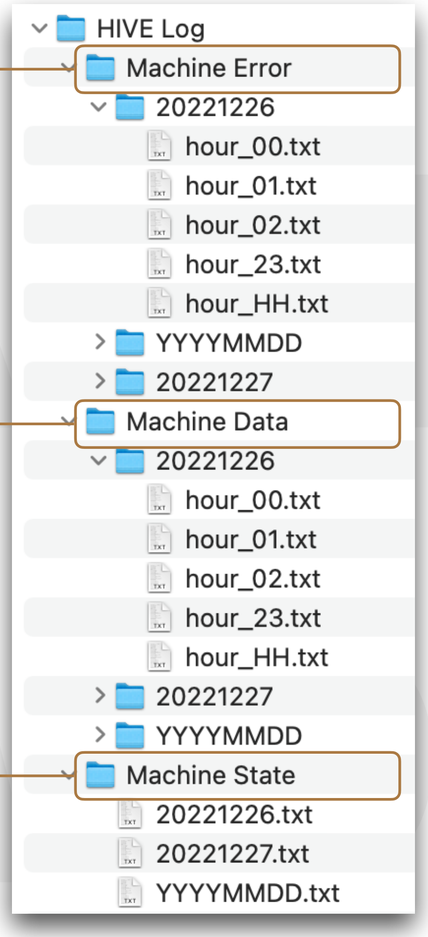
Machine error sorts by date folders. Naming rule: **YYYYMMDD**. For example 20221227

Each hourly log is recorded in one log file. Naming rule: **hour_HH**. For example hour_00, meaning this file is the machine log from 00:00 to 01:00

Machine State stores by daily. Each daily log is recorded in one log file. Naming rule: **YYYYMMDD**. For example 20221226

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HIVE log includes **Machine State**, **Machine Error** and **Machine Data**. Each stored in separated folder



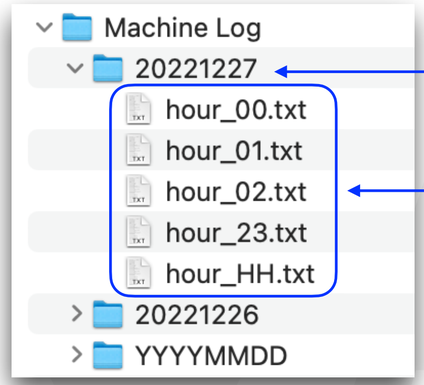
An example for HIVE log

```
00:03:54-->Http开始发送数据:{
  "unit_sn": "GX730130J2M1TN7AX",
  "serials": {
    "top_case": "",
    "bottom_case": ""
  },
  "pass": "true",
  "input_time": "2023-01-04T00:03:31.21+0800",
  "output_time": "2023-01-04T00:03:54.63+0800",
  "data": {
    "CT": "23",
    "sw_version": "JQS_5.1.0.1_221108_POR",
    "MS_SHA1": "",
    "CD_SHA1": "",
    "VS_SHA1": "",
    "VL1_SHA1": "",
    "RS_SHA1": ""
  }
}
00:03:54-->Http文本转换完成
00:03:54-->Http数据流写入完成
00:03:54-->Http接收返回值:{"ErrorCode":null,"ErrorText":null,"Status":"Success
machinedata logged"}

00:04:13-->Http开始发送数据:{
  "unit_sn": "GX730130LRM1TN7AU",
  "serials": {
    "top_case": "",
    "bottom_case": ""
  },
  "pass": "true",
  "input_time": "2023-01-04T00:03:50.27+0800",
  "output_time": "2023-01-04T00:04:13.56+0800",
  "data": {
    "CT": "23",
    "sw_version": "JQS_5.1.0.1_221108_POR",
    "MS_SHA1": "",
    "CD_SHA1": "",
    "VS_SHA1": "",
    "VL1_SHA1": "",
    "RS_SHA1": ""
  }
}
00:04:13-->Http文本转换完成
00:04:13-->Http数据流写入完成
00:04:13-->Http接收返回值:{"ErrorCode":null,"ErrorText":null,"Status":"Success
machinedata logged"}
```

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Machine Log



Machine log sorts by date folders. Naming rule: **YYYYMMDD**. For example 20221227

Each hourly log is recorded in one log file. Naming rule: **hour_HH**. For example hour_00, meaning this file is the machine log from 00:00 to 01:00

Machine Log Example

2022/12/23 19:07:00:104 User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 User login
2022/12/23 19:08:03:803 User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 New unit input detected by sensor A-1-2
2022/12/23 19:08:03:813 User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Start scanning process xxxx

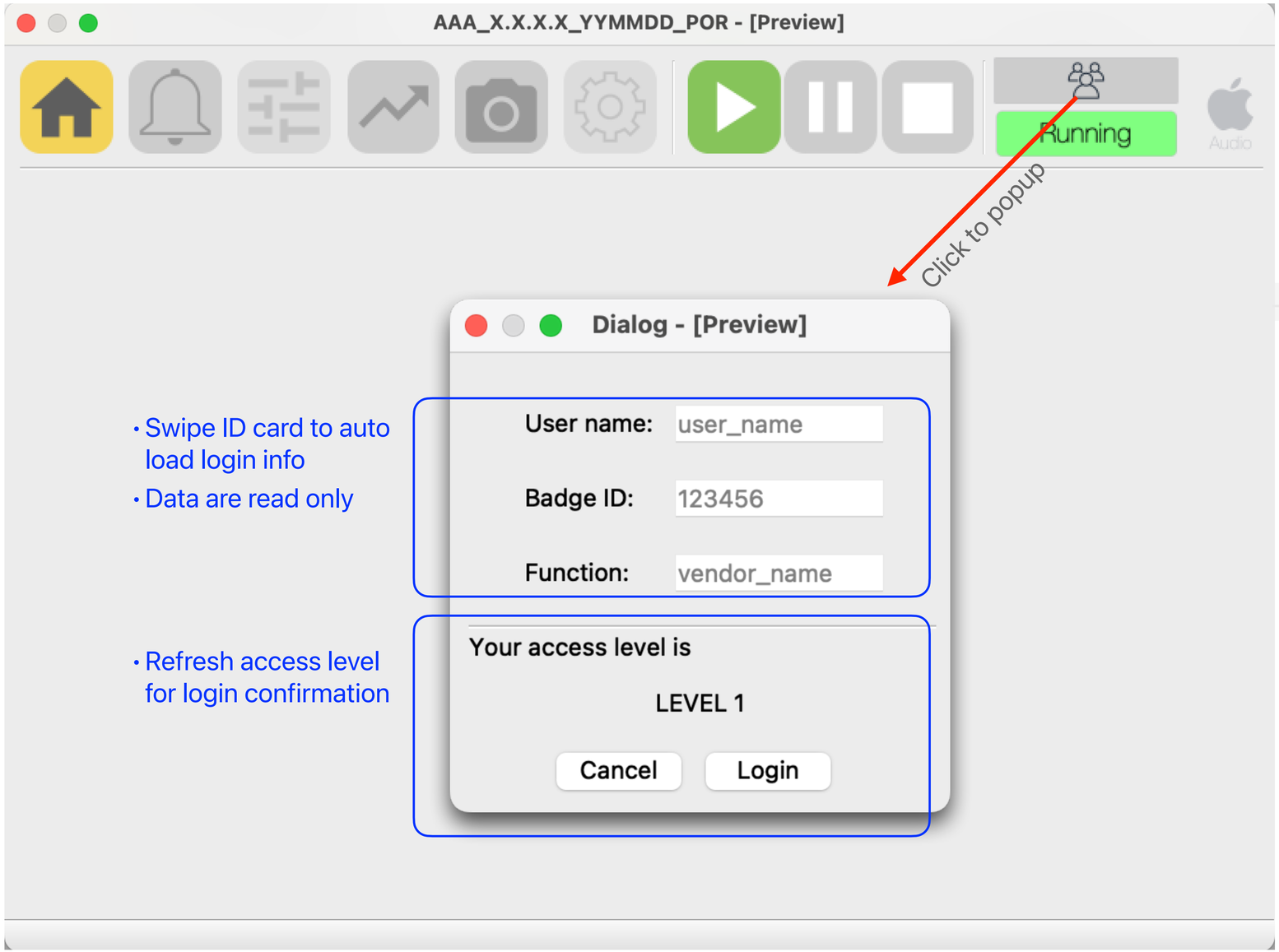
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 XXXX
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 XXXX
... ...
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Scanning completed. Unit SN XXXXX. Fixture SN XXXXX
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Step1
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Step2
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 New unit input detected by sensor A-1-2
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Step3
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Interact with MES
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 XXXX
.....
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Interact with Insight
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 XXXX
.....
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 Interact with HIVE
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 XXXX
.....
YYYY/MM/DD HH:MM:SS:sss User:San.Zhang Function:ABC_OSS UID:1234567890 Authorize:Wu.Wang Function: ICT_DFX UID:0987654321 [Error/Alert] Error description
.....
YYYY/MM/DD HH:MM:SS:sss User:null Function:null UID:null Authorize:null Function:null UID:null User logout
YYYY/MM/DD HH:MM:SS:sss User:null Function:null UID:null Authorize:null Function:null UID:null XXXX
.....
YYYY/MM/DD HH:MM:SS:sss User:null Function:null UID:null Authorize:null Function:null UID:null Interact with HIVE
YYYY/MM/DD HH:MM:SS:sss User:null Function:null UID:null Authorize:null Function:null UID:null XXXX
.....
YYYY/MM/DD HH:MM:SS:sss User:null Function:null UID:null Authorize:null Function:null UID:null [Error/Alert] Error description
.....

- Start new row for now log event with timestamp
- Timestamp format YYY/MM/DD HH:MM:SS:mmm. For example 2023/05/12 20:04:08:809
- Keep a space between timestamp and log description
- Record login user name, function name, u_id log file.
- User NULL as u_id value when there are no logins

Key Requirements:

- Must include SN information use 000000000 if no SN is scanned or scanning is disabled
- Must include main operation steps of each machine. For example, dispenser's dispensing actions. Laser machine's soldering actions. Auto PnP's vision detection and assembly actions. Account login and log out. etc.
- Must include records regarding to interaction with MES/Insight/HIVE. If MES/Insight/HIVE is disabled, must show MES/Insight/HIVE is disconnected in machine log
- For errors reported by machine, log description starts with [Error/Alert]. Then followed by error descriptions

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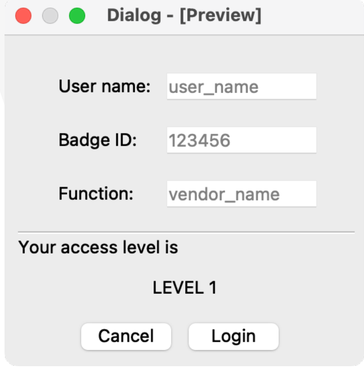
- Swipe ID card to auto load login info
- Data are read only
- Refresh access level for login confirmation

Key Requirements

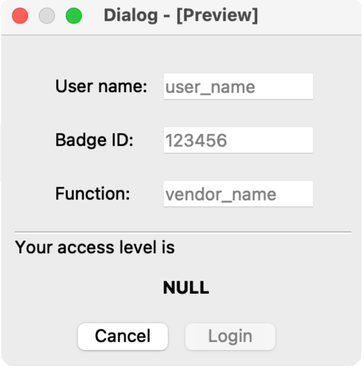
- Must follow software version name and upgrade rule
- Version code hard coded in codes, enable changing through config file is not allowed
- All modules are ***read only*** when no login

No login		
L1	Level 1	User ID is logged. Normal production access
L2	Level 2	Engineer ID is logged. Universal change access
L3	Level 3	Admin ID is logged. All actions must be recorded by log

Normal access



No access



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S1.2-Access

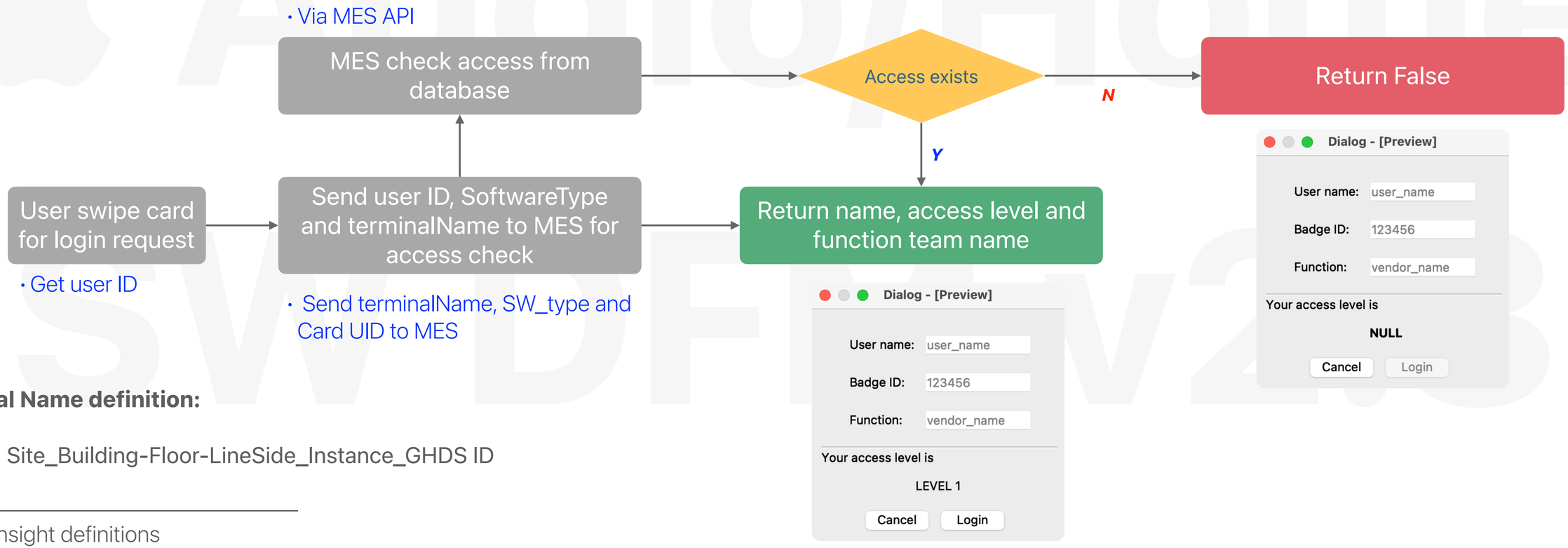


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Sample Access Database

Site	Project	Station ID	Terminal Name	Name	Access Level	SoftwareType	Card UID	Function
ITKS	H37	FT200	ITKS_E03-4FT-01B_1_STATION59	Name 1	3	Main	123456	Apple AME
ITKS	H37	FT200	ITKS_E03-4FT-01B_1_STATION59	Name 2	2	Main	123	ICT DFx
ITKS	H37	FT200	ITKS_E03-4FT-01B_1_STATION59	Name 1	3	Vision	123456	Apple AME
ITKS	H37	FT200	ITKS_E03-4FT-01B_1_STATION59	Name 2	2	Main	123	ICT DFx

Machine -> MES
MES -> Machine



Terminal Name definition:

Site_Building-Floor-LineSide_Instance_GHDS ID

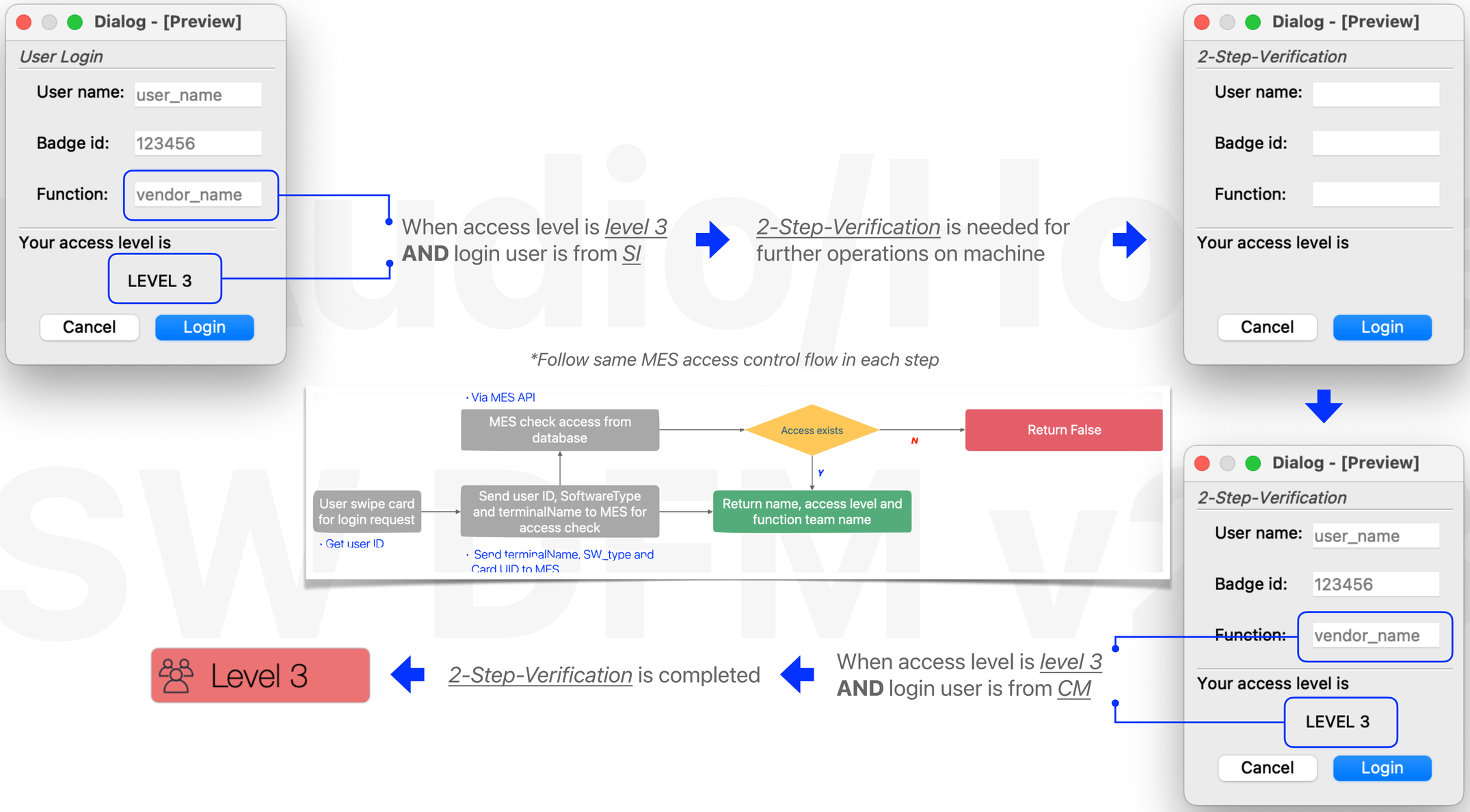
*Follow Insight definitions

*All capitalized

Apple Confidential

S1.2.1-2 Step Verification for SI Level 3 Access

Audio/Home
SW DFM v2.8



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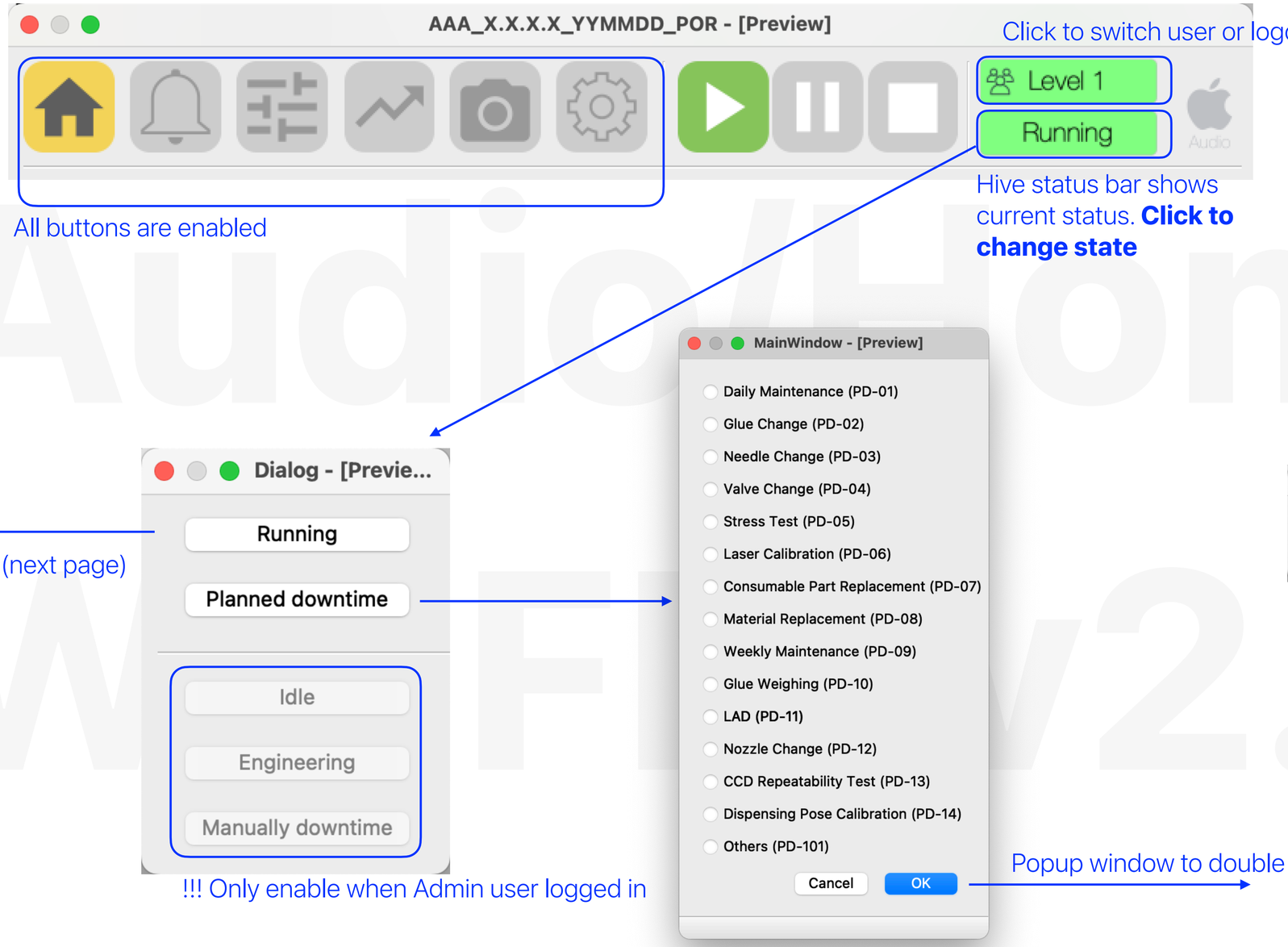
S1.3-Loged in

Level 1

Level 2

Level 3

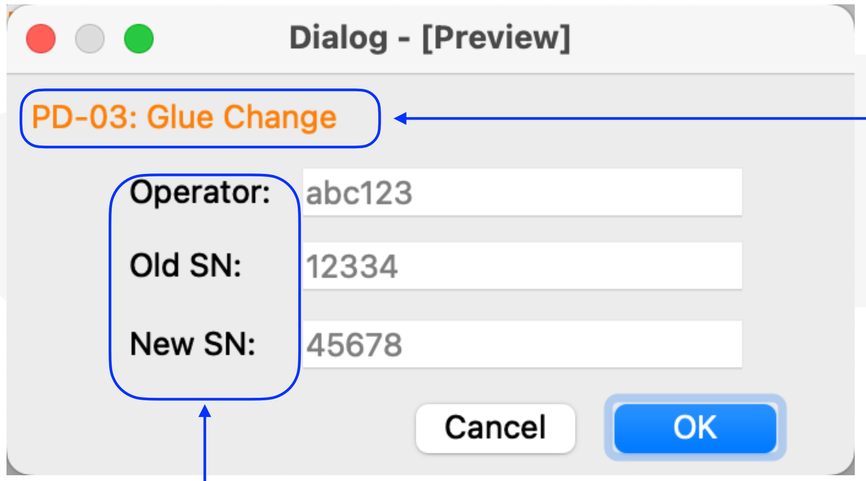
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When entering **ANY state by manually**, popup window for filling operator/SN info and record to HIVE

For planned downtime that are related with a physical SN

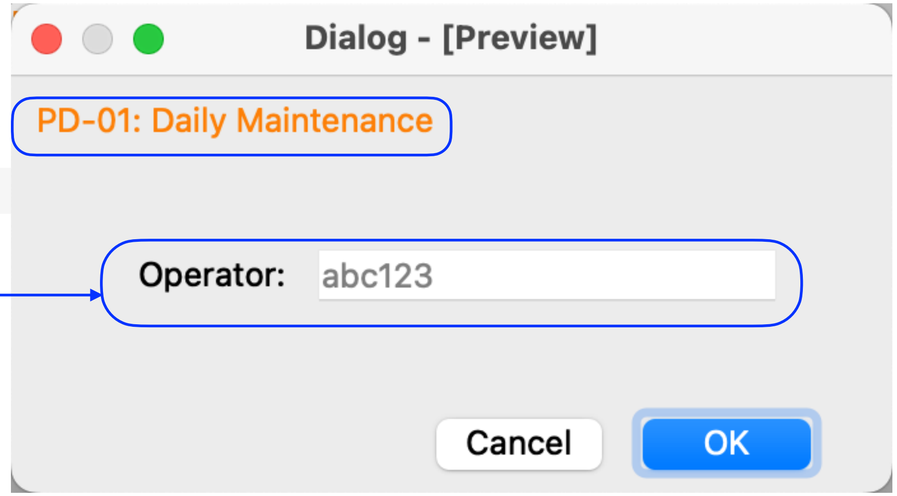


- Swipe ID card to load operator ID
- Old SN will show as recorded
- Use scanner to auto fill in new SN. Manually key in is **NOT** allowed

Type of planned downtime

Swipe badge card to auto read UID. Manually key in is **NOT** allowed

For planned/unplanned downtime that are NOT related with a physical SN



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S1.3.1-Home

Audio/Home
SW DFM v2.8

Audio/Home TechOps | SW DFM



Click space areas to jump to error tab

Click space areas to jump to data tab

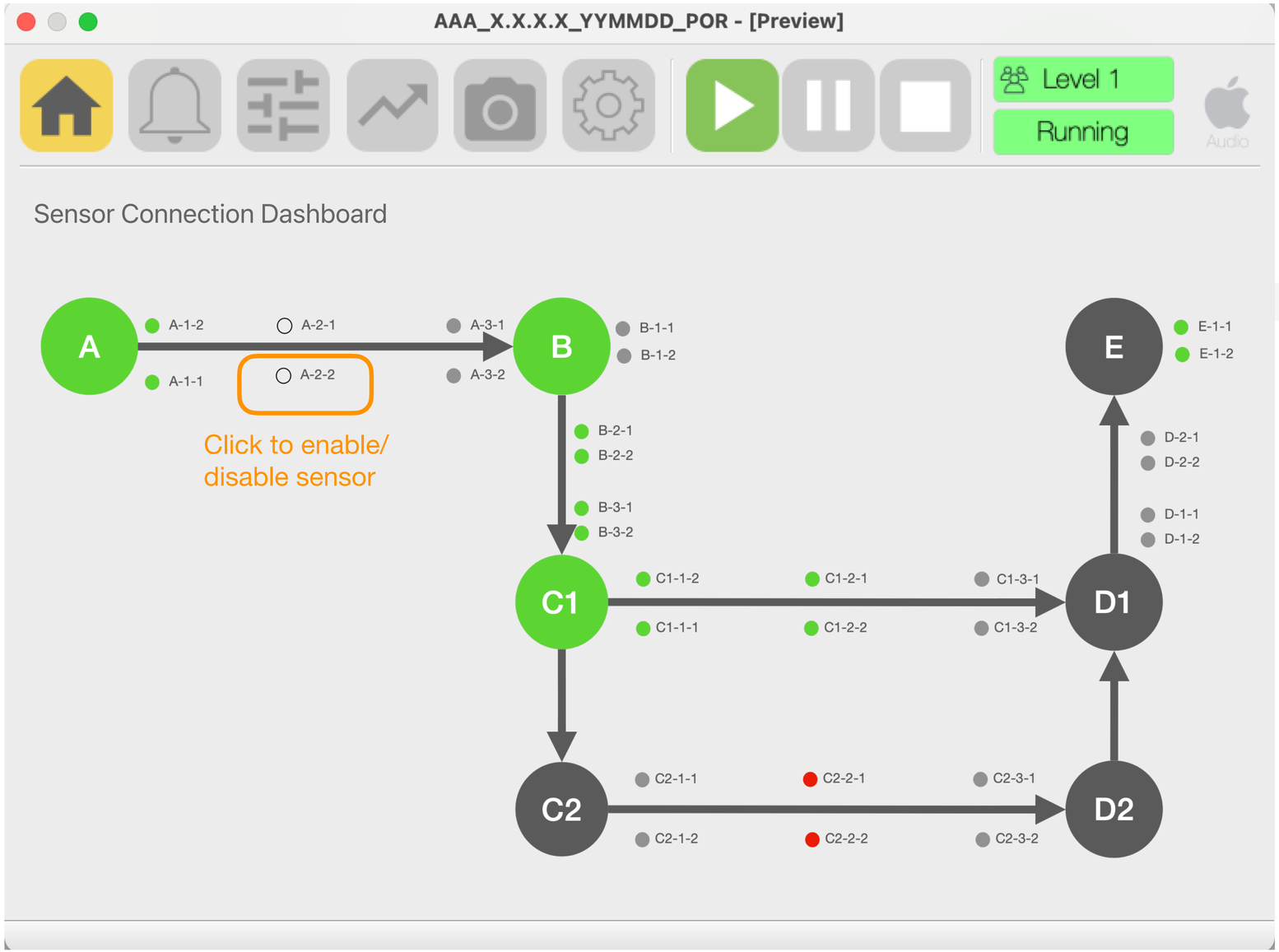
Show sensor connection button after login



- Main windows under HOME tab to show the dashboard defined on the left

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Audio/Home TechOps | SW DFM



Key Requirements

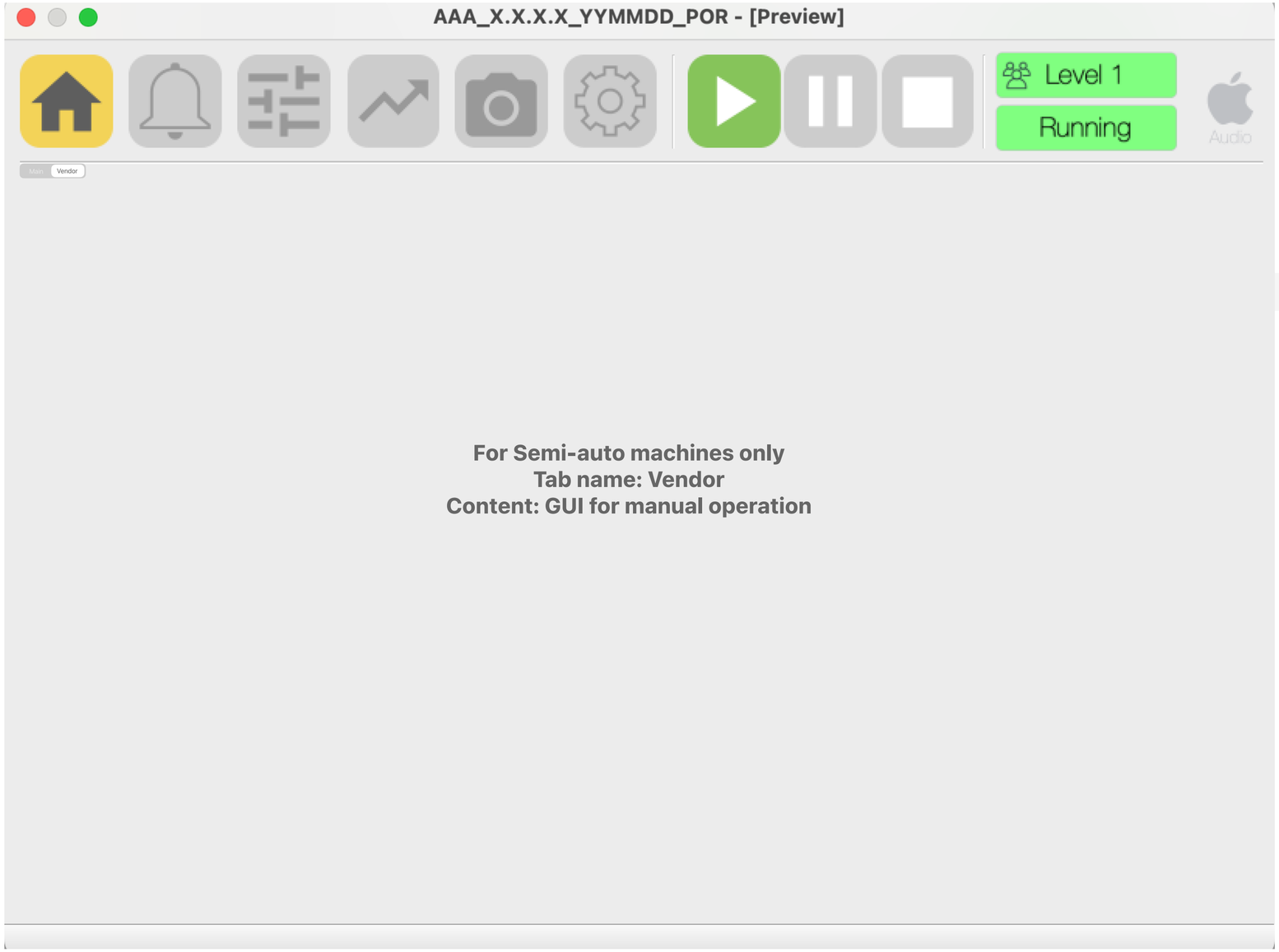
- Sensor connection dashboard to show sensor/station status lively
- Level 1: View only
- Level 2, 3: Full access

- Only non-critical sensor can be disabled by manually
  - Critical: Sensor disablement may cause safety issue, for example the sensor to detect the location of the carrier/fixture
  - Non-critical: Sensor can be disable to quickly resume production and no safety or collision concern after disablement

A	Unit at location A
B	No units at location B
A-3-2	Sensor A-3-2 is idle
C1-1-2	Sensor C1-1-2 is activated
A-2-2	Sensor A-2-2 is manually disabled
C2-2-1	Sensor C2-2-1 signal lost

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Audio/Home TechOps | SW DFM



Main Vendor

- Vendor window for semi-auto stations only(require OP loading/ scanning)

Home
v2.8

S1.3.2-Alert

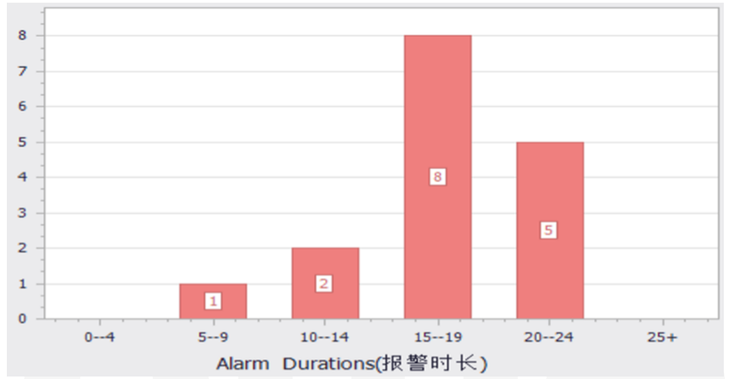
Audio/Home
SW DFM v2.8



1/2 height

Date	Start Time	End Time	Time in State (Min)	Error Code	Error Message	Error Detail
YYYY-mm-dd	YYYY-mm-dd HH:MM:ss.SS		Ongoing			
YYYY-mm-dd	YYYY-mm-dd HH:MM:ss.SS	YYYY/mm/dd HH:MM:ss.SSS	13	Follow error code definition	Follow error type definition	Show original errors reported by machine
2022-12-26	2022-12-26 13:59:05.061	2022-12-26 13:59:16.061	11	Follow error code definition	Safety Error	Show original errors reported by machine

1/2 height



Key Requirements

- X axis: statistics in minutes, Example, < 5 minutes are classified as (0-4)...
- Statistics of times

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Key Requirements

- When machine alarms, it automatically switches to the alarm page and pops up a window. HIVE state updates to downtime(state 5) with corresponding messages sent to HIVE endpoint
- Once confirm window is clicked, pop up window for card scan

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Key Requirements

- When machine alarms, it automatically switches to the alarm page and pops up a window. HIVE state updates to downtime(state 5) with corresponding messages sent to HIVE endpoint
- Once confirm window is clicked, pop up window for card scan
- After card is swiped, and OK button is clicked, update payload with

  state_change_time=current timestamp
  previous state=5
  error_message=actual error message
  error_detail=actual error detail
  u_id=actual card id

and resend as HIVE machine state

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S1.3.3-Config

Audio/Home
SW DFM v2.8



!!! Level 1: View only for this tab

Home v2.8



!!! Level 2 can change parts of the machine configs. (Non-critical only)

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!!! Only enable when Admin user logged in

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S1.3.4-Data

Audio/Home
SW DFM v2.8

Audio/Home TechOps | SW DFM



Data	Shift	Unit SN	Component SN (Add all that are linked)	Start Time	End Time	Pass	CT/s	HIVE State	Tossing
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX					0
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	TRUE	X	1	1
YYYY/MM/DD	NS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	FALSE	X	1	1
2023/6/12	DS	FX49RT9527	N/A	2023/6/12 12:59:47.492	2023/6/12 12:59:58.124	TRUE	12.89	1	2

Key Requirements

- Filter time
- IO/Tossing
- Data Export

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Audio/Home TechOps | SW DFM



Data	Shift	Unit SN	Component SN (Add all that are	Start Time	End Time	Pass	CT/s	HIVE State	Tossing
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX					0
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	TRUE	X	1	1
YYYY/MM/DD	NS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	FALSE	X	1	1
2023/6/12	DS	FX49RT9527	N/A	2023/6/12 12:59:47.492	2023/6/12 12:59:58.124	TRUE	12.89	1	2

Key Requirements

- Filter time
- IO/Tossing
- Data Export

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Audio/Home TechOps | SW DFM



Data	Shift	Unit SN	Component SN (Add all that are linked)	Start Time	End Time	Pass	CT/s	HIVE State	Tossing
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX					0
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	TRUE	X	1	1
YYYY/MM/DD	NS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	FALSE	X	1	1
2023/6/12	DS	FX49RT9527	N/A	2023/6/12 12:59:47.492	2023/6/12 12:59:58.124	TRUE	12.89	1	2

Key Requirements

- Filter time
- IO/Tossing
- Data Export

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Audio/Home TechOps | SW DFM



Data	Shift	Unit SN	Component SN (Add all that are	Start Time	End Time	Pass	CT/s	HIVE State	Tossing
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX					0
YYYY/MM/DD	DS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	TRUE	X	1	1
YYYY/MM/DD	NS	Xxx	HH:MM:SS:XX	HH:MM:SS:XX	HH:MM:SS:XX	FALSE	X	1	1
2023/6/12	DS	FX49RT9527	N/A	2023/6/12 12:59:47.492	2023/6/12 12:59:58.124	TRUE	12.89	1	2

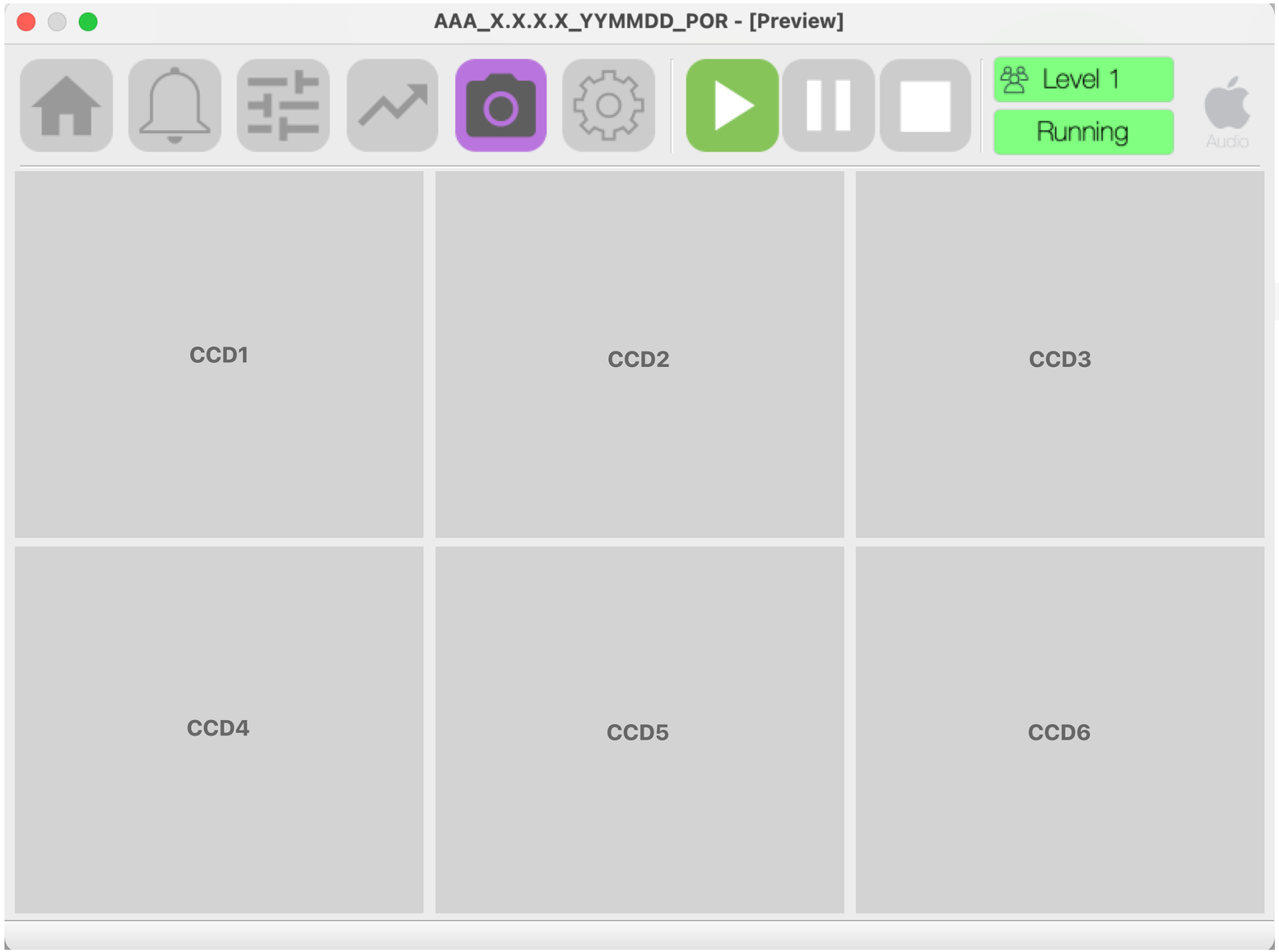
Key Requirements

- Filter time
- IO/Tossing
- Data Export

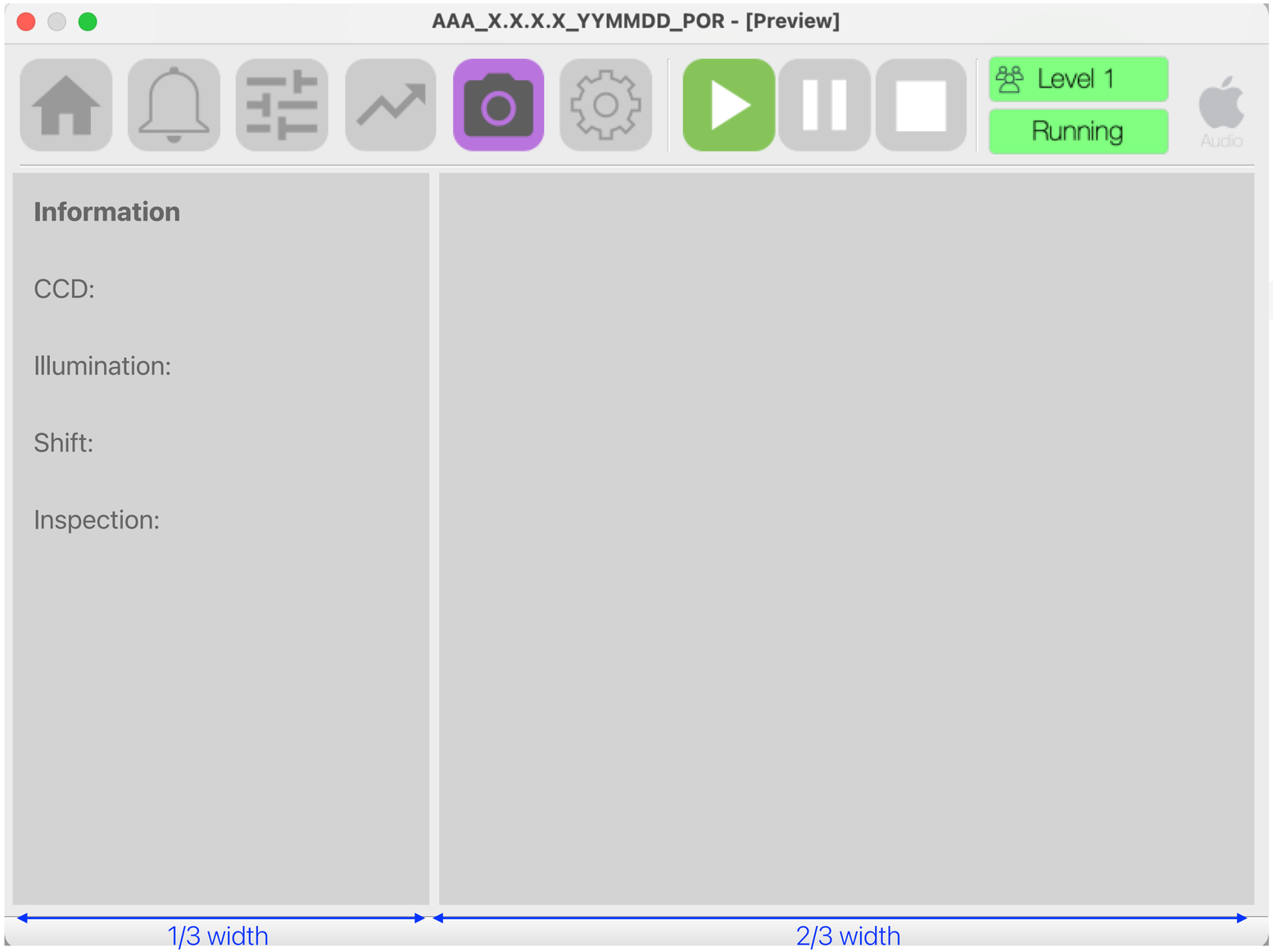
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S1.3.5-Vision

Audio/Home
SW DFM v2.8


AAA_X.X.X.X_YYMMDD_POR - [Preview]
Level 1
Running
Audio
CCD1
CCD2
CCD3
CCD4
CCD5
CCD6

Home v2.8


AAA_X.X.X.X_YYMMDD_POR - [Preview]
Level 1
Running
Audio
Information
CCD:
Illumination:
Shift:
Inspection:
1/3 width
2/3 width

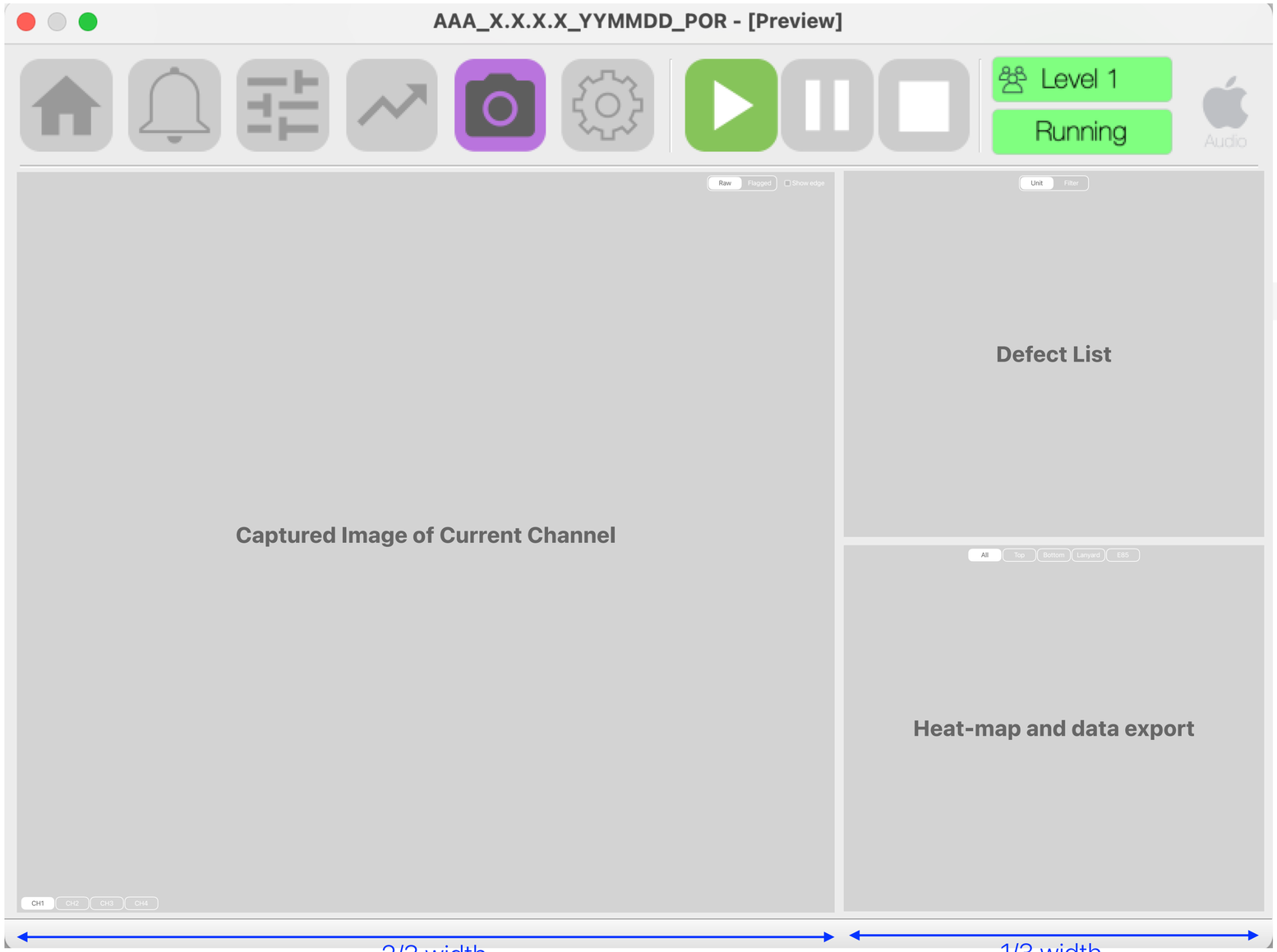
Example


Information
CCD: Image1,1
Unit: pcs
Angle: -0.123
X_Gap: 0.234
Y_Gap: 0.165
Left_Gap: 0.165
Delta_XGap: 0.165
Delta_YGap: 0.165
Delta_Left_Gap: 0.165
T1_Brightness: 0.165
T1_Exposure: 8ms

Vision Page for AOI

Audio/Home
SW DFM v2.8

Audio/Home TechOps | SW DFM

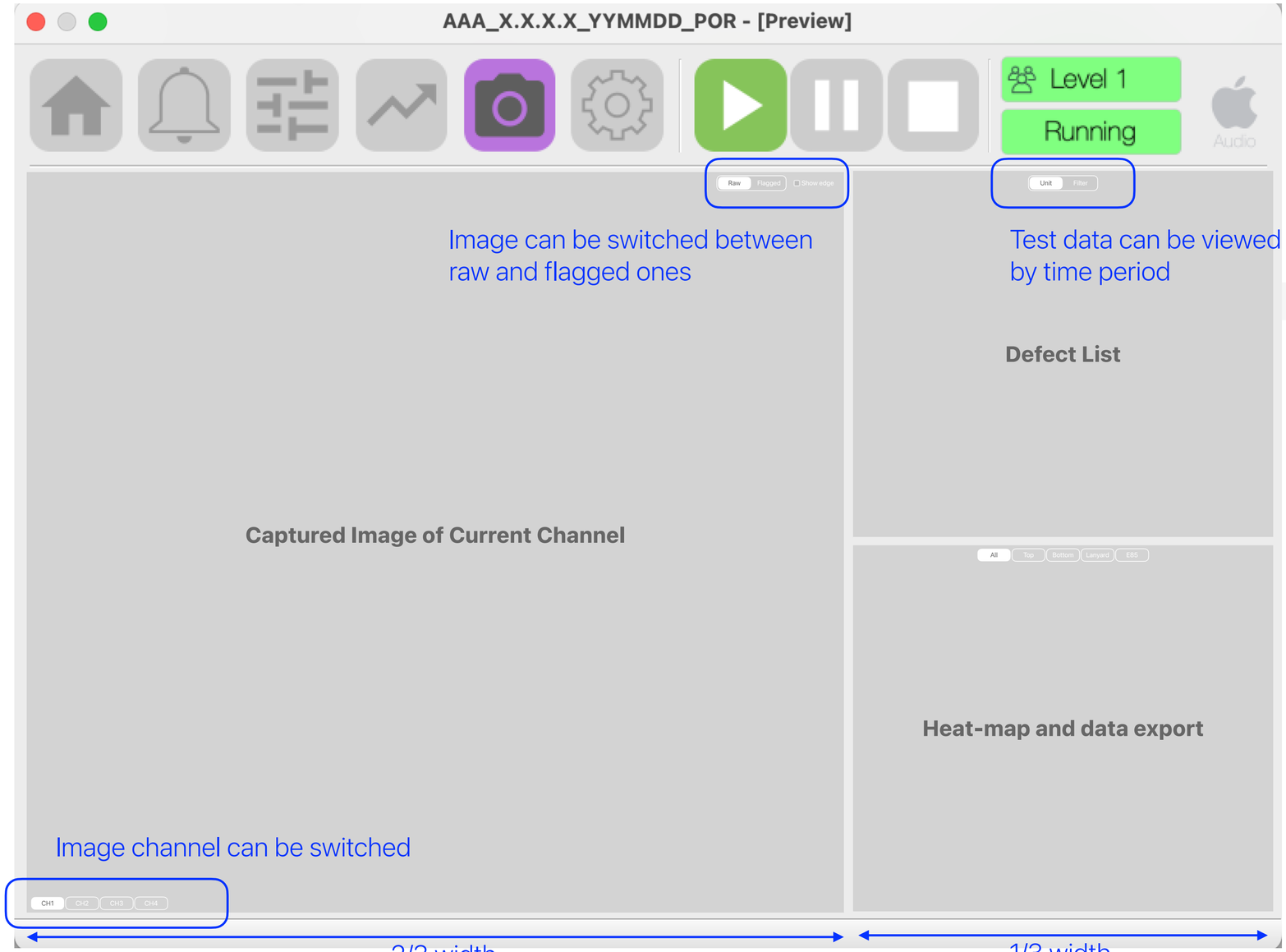


Key Requirements

- Image channel can be switched
- Image can be switch between raw and flagged ones
- Heat-map can be viewed by unit or filtered by time period

Home v2.8

Audio/Home TechOps | SW DFM



Key Requirements

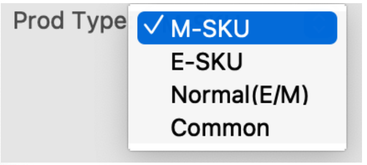
- Image channel can be switched
- Image can be switch between raw and flagged ones
- Heat-map can be viewed by unit or filtered by time period

Audio/Home
SW DFM v2.8

S1.3.6-Setting

Audio/Home
SW DFM v2.8





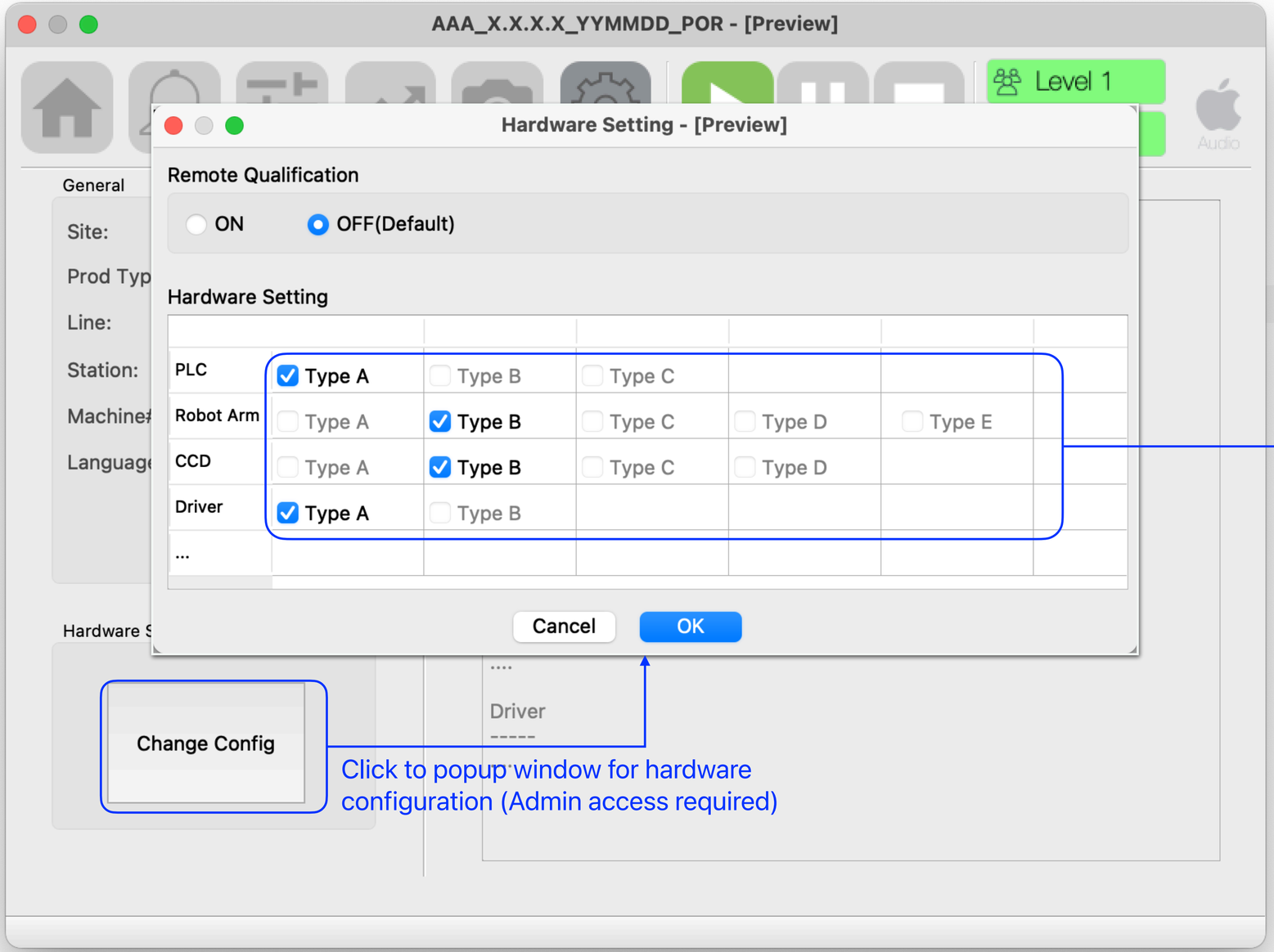
- For Level 1 and Level 2 users access, product type will show as read-only. Any edited is NOT allowed
- Change product type will need Level 3 access

Key Requirements

- Link to station card on Home page

- Cross QPL/Line/Site software implementation will not allow re-compile software
- Use the config button to config hardware of each machine(if a machine has different hardware)

- Change site, line, station and machine number will require admin access
- Change language will not require admin access
- Any change should be selected through the menu bar. Manually key-in is NOT allowed.



1. Select to config hardware type
2. Click ***Load Config*** to save

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1. Once saved, the actual config of this machine will show here
2. Software backend will auto switch code block according

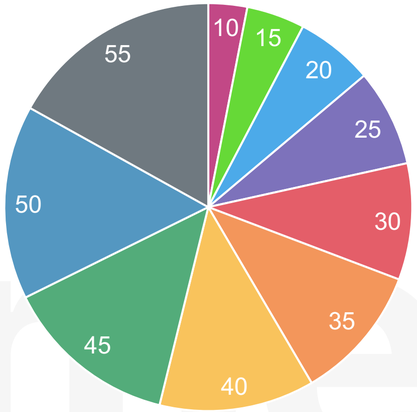
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S1.4-Color Code Definition

Audio/Home
SW DFM v2.8

Audio/Home TechOps | SW DFM - Color Map

State Name	Running	Idle	Engineering	Planned downtime	Downtime
Hex	#D3EB73	#EBFFFE	#CCABD8	#FFD7D4	#EB7373
RGB	R: 211 G: 235 B: 115	R: 235 G: 255 B: 254	R: 204 G: 171 B: 216	R: 255 G: 215 B: 212	R: 235 G: 115 B: 115



Unit SN

Background R: 0 G: 249 B: 0

Unit SN

Background R: 236 G: 93 B: 87

STN01 #1

Site: XXXX **Vendor:** AAA
SW Version: AAA_X.X.X.X_YYMMDD_POR
Main SW Path: *C:/path/path/name.exe*
MS_Hash: 872d5c91f16108bd02fdc16a1e60752385c161f6
HIVE Connected MES Connected PDCA Disconnected
Machine Log HIVE Log

Background R: 255 G: 143 B: 0

Normal(E/M)

Background R: 57 G: 130 B: 81

M-SKU

Background R: 121 G: 132 B: 231

E-SKU

Background R: 246 G: 174 B: 66

STATION ID

Text R: 204 G: 204 B: 204

Downtime Statistics Top 10

Top10	R	G	B	Hex
Top 1	111	121	128	6F7980
Top 2	84	151	193	5497C1
Top 3	83	172	122	53AC7A
Top 4	248	195	92	F8C35C
Top 5	243	150	91	F3965B
Top 6	228	94	105	E45E69
Top 7	125	114	187	7D72BB
Top 8	76	170	233	4CAAE9
Top 9	102	217	56	66D938
Top 10	194	72	134	C24886

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S2-HIVE and SW Backend Requirements

Audio/Home
SW DFM v2.8

S2.1-HIVE Basic

Audio/Home
SW DFM v2.8

HIVE State Definition					
State Name	Running	Idle	Engineering	Planned downtime	Downtime
State Number	1	2	3	4	5
Definition	Machine is working normally on processing units	Machine is working normally but no units being processed	Machine needs unplanned units processing(e.g. skip MES)	Planned machine maintenance within scheduled period and frequency	Machine down due to unexpected error or wrong operation
Trigger Rules	When there is a unit coming	No units coming in 90 seconds under running or engineering status	DOE that requires to skip MES	Daily/Weekly maintenance.	Machine error or alarm. OSS/OP/EQ start to deal with downtime (Auto change to state 5)
	When state Engineering/Planned Downtime/Downtime ends		All other production modes without pass MES	Glue/valve or similar periodic actions(e.g. material change, nozzle replacement, etc.)	All unplanned actions/upgrade tasks (Manually change to state 5)
	Machine need to be powered off (Should wait for 90s for machine turning to idle before power-off)			All other kinds of planned (need Apple approval) tasks	Units try run after downtime debug (skip MES)
	Enable MES	***Force to skip MES. No ways to enable MES***			

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RULE #1:

HIVE is line critical: HIVE comes with a problem will cause machine downtimes (Need Apple AME approval if need to temporarily disable HIVE).

RULE #2:

MES is forced to be disabled under state 2(idle), 3(engineering), 4(planned downtime) and 5(downtime). No ways to be enabled by any means. MES is automatically enabled under state 1(running).

RULE #3:

When no units are processed within 90s, machine will be automatically switched from state 1(running) or 3(engineering) to state 2(idle)

RULE #4:

State 3 is not allowed to be used from MP. Only available during NPI builds for DOE run without enabling MES

RULE #5:

Machine only update state data when state/error changes. Periodically state change is prohibited

RULE #6:

Reset machine or press running/idle button on GUI will not change HIVE state

Audio/Home SW DFM v2.8

Audio/Home TechOps | SW DFM - HIVE Error Messages - Planned Downtime

	ErrorCode	ErrorType	Explain	Error Detail
Planned Downtime	PD-01	Daily Maintenance	All routine maintenance work(e.g. routine check before production)	
	PD-02	Glue Change	Normal glue change actions or alarm when glue runs out. Exclude any unplanned changes. Scan to input new cartridge SN when entering this state	GantryID_glue type. e.g. front_AB glue, rear_HM glue
	PD-03	Needle Change	Normal needle change actions or alarm. Exclude any unplanned changes	
	PD-04	Valve Change	Normal valve change actions or alarm. Exclude any unplanned changes	
	PD-05	Stress Test	When running stress test without passing MES	
	PD-06	Laser Calibration	Planned laser calibration actions. Exclude any non-planned calibrations	
	PD-07	Consumable Part Replacement	Other consumable part replacement actions or alarm, such as replacing wiper, tin ball. Exclude needle, valve, glue or material change	Specify action details by manual input/selection when choosing this error
	PD-08	Material Replacement	All planned material change actions	
	PD-09	Weekly Maintenance	When machines are going to idle for more than half a day as scheduled(e.g. entering weekend without production plan)	
	PD-10	Glue Weighing	Regular glue weighing	
	PD-11	LAD	Special mode for uploading data in audit mode for LAD	Auto generate base on LAD test mode. TestSeriesID_CPK, TestSeriesID_GRR, etc
	PD-12	Nozzdle Change	Normal nozzdle change actions or alarm. Exclude any non-planned changes	
	PD-13	CCD Repeatability Testing	Vision system repeatability tests: Static test (parts & CCD are held static), Dynamic test 1 (parts moves_dynamic, CCD is held static), Dynamic test 2 (parts is held static, CCD moves_dynamic).	Auto generates similar to the LAD test mode. TestSeriesID_CPK, TestSeriesID_GRR, etc.
	PD-14	Dispensing Pose Calibration		
	PD-101	Others	All other planned actions not listed above	Specify action details by manual input/selection when choosing this error

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Audio/Home TechOps | SW DFM - HIVE Error Messages - Unplanned Downtime

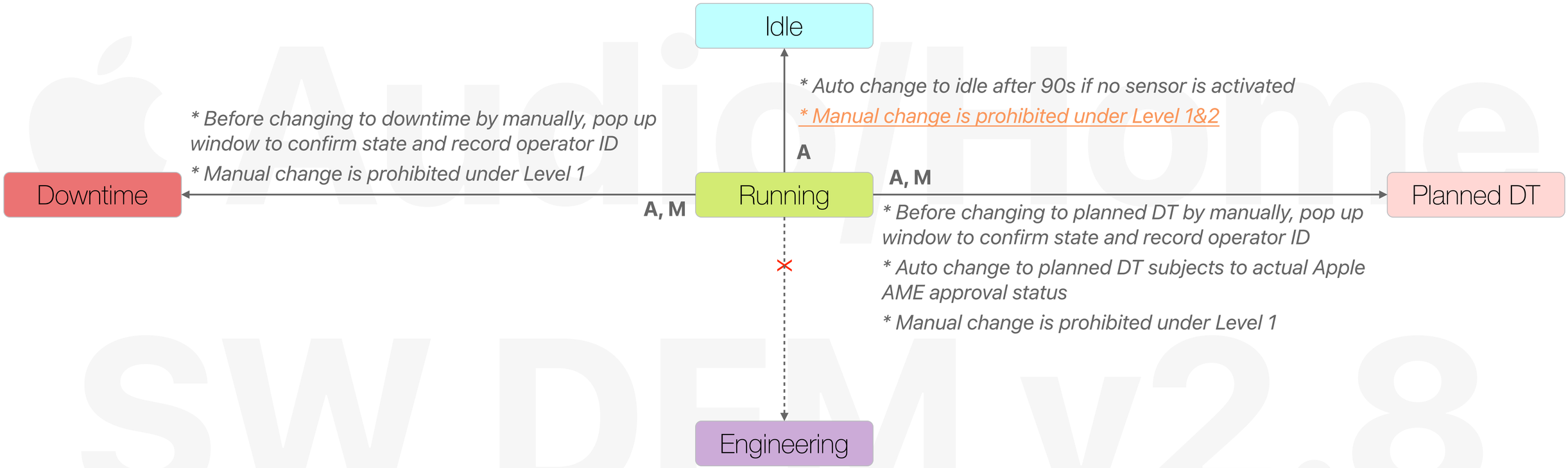
	ErrorCode	ErrorType	Explain	Error Detail
Unplanned Downtime	C-	Cylinder Error		
	S-	Sensor Error		Specify the sensor name and sensor location. E.g. Sensor A-1-1 alarms
	N-	SW Error		
	V-	Vision Error		Specify CCD ID in error detail
	F-	Safety Error		
	T-	Scanning Error		
	M-	Motion Error		
	U-	Vacuum Error		
	W-	Screwing Error		
	L-	Laser Error		
	G-	Glue Error	e.g. Glue path file missing. Exclude glue runs out	
	P-	PDI Error	Post Dispense Inspection. Use this type when post inspection fail	
	Y-	Manually Downtime		Specify details
	A-	Open Time Control	For between station open time control	STN# downtime: Error Type/WIP control
	X-	MES Error		
	O-	Others		Specify action details by manual input/selection when choosing this error

Audio/Home SW DFM v2.8

Audio/Home TechOps | SW DFM

A: Auto Switch
M: Manual Switch

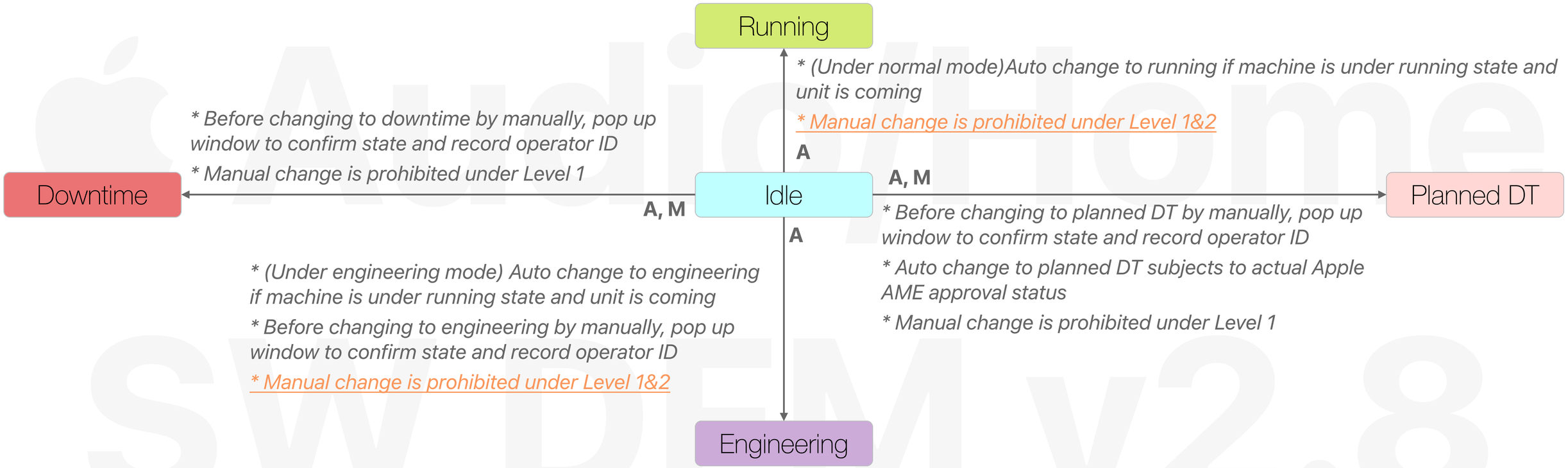
Go back to the contents page



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A: Auto Switch
M: Manual Switch

Go back to the contents page

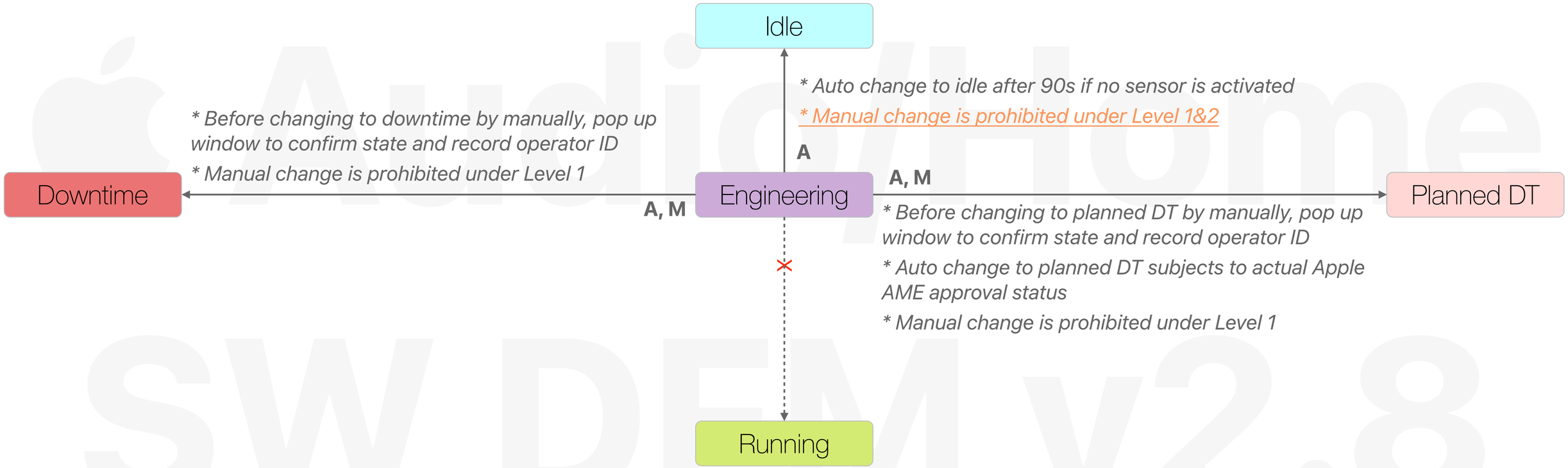


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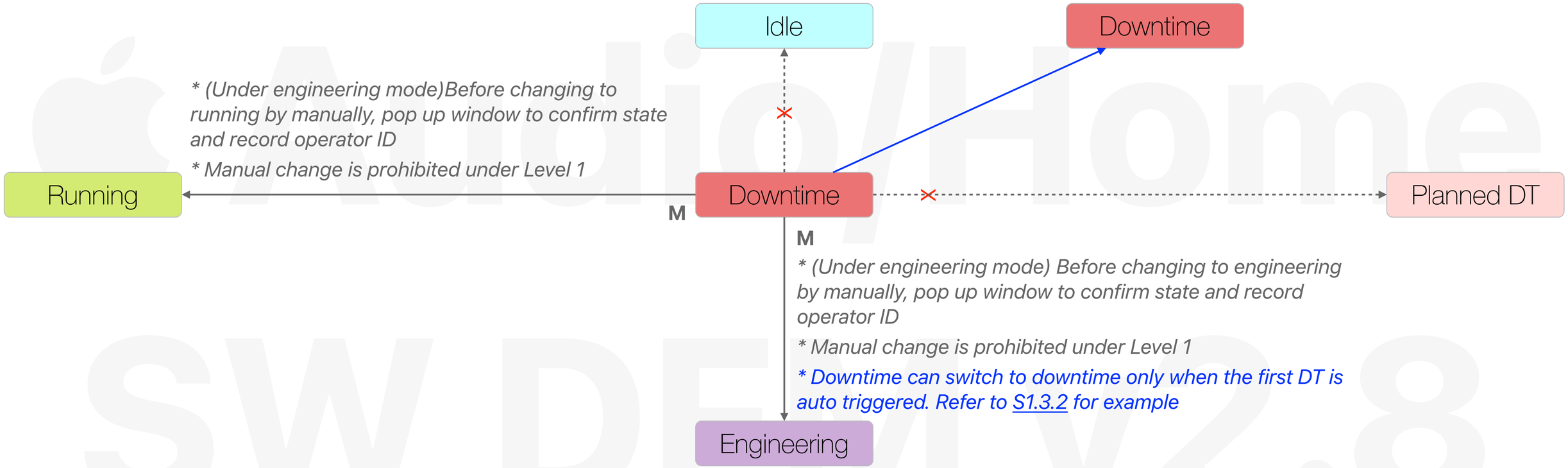
Audio/Home TechOps | SW DFM

A: Auto Switch
M: Manual Switch

Go back to the contents page



Audio/Home SW DFM v2.8

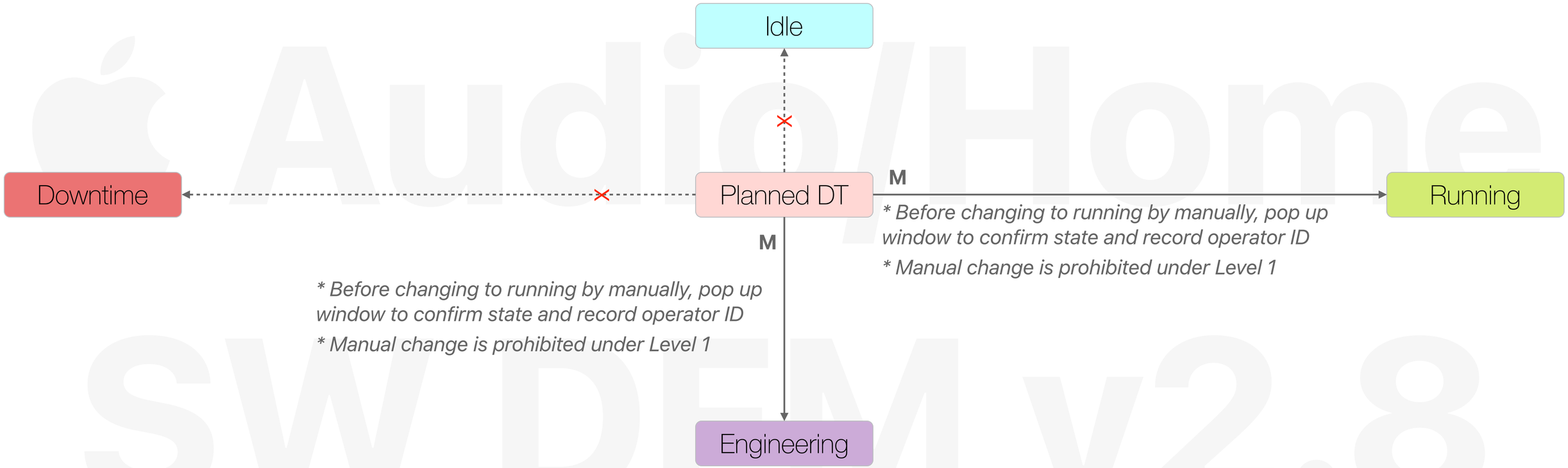


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Audio/Home TechOps | SW DFM

A: Auto Switch
M: Manual Switch

Go back to the contents page



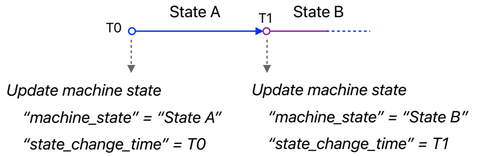
Audio/Home
SW DFM v2.8

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S2.2-HIVE Data Reporting

Audio/Home
SW DFM v2.8

Machine State



1: Running

```
{
    "machine_state": 1,
    "state_change_time": "2022-02-01T12:59:05.06+0800",
    "data": {
        "previous state": 2,
        "sw_version": "xxxxx",
        "MS_SHA1": "xxxxx" ,
        "VS_SHA1": "xxxxx" ,
        "u_id": "xxxxx"
        }
}
```

2: Idle

```
{
    "machine_state": 2,
    "state_change_time": "2022-02-01T12:59:05.06+0800",
    "data": {
        "previous state": 1,
        "sw_version": "xxxxx",
        "MS_SHA1": "xxxxx" ,
        "VS_SHA1": "xxxxx" ,
        "u_id": "xxxxx"
        }
}
```

3: Engineering

```
{
    "machine_state": 3,
    "state_change_time": "2022-02-01T12:59:05.06+0800",
    "data": {
        "previous state": 1,
        "sw_version": "xxxxx",
        "MS_SHA1": "xxxxx",
        "VS_SHA1": "xxxxx",
        "u_id": "xxxxx"
        }
}
```

4: Planned downtime

```
{
    "machine_state": 4,
    "state_change_time": "2022-02-01T12:59:05.06+0800",
    "data": {
        "previous state": 1,
        "sw_version": "xxxxx",
        "MS_SHA1": "xxxxx" ,
        "VS_SHA1": "xxxxx" ,
        "error_message": "Sensor Error",
        "error_detail": "xxxxx" ,
        "u_id": "xxxxx" ,
        "old_sn": "xxxxx" ,
        "new_sn": "xxxxx"
        }
}
```

5: Downtime

```
{
    "machine_state": 5,
    "state_change_time": "2022-02-01T12:59:05.06+0800",
    "data": {
        "previous state": 1,
        "sw_version": "xxxxx",
        "MS_SHA1": "xxxxx" ,
        "VS_SHA1": "xxxxx" ,
        "error_message": "Sensor Error",
        "error_detail": "xxxxx",
        "u_id": "xxxxx"
        }
}
```

Accurate to millisecond

Use actual time zone. +0800 for CN. +0700 for VN

Follow key naming rule to upload all needed hash key value

Upload error message and error detail for planned downtime and downtime

Follow error type definition, strings not listed in definition is not allowed

*Always require to swipe operator's (OSS/FE/EQ etc.) badge and upload to hive when a state is switched by manually: "u_id": "xxxxx"

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Machine State

	Main SW	Vision SW	Laser SW	Robot SW	IDE	Config File	Vision Template 1 (Localization)	Vision Template 2 (Localization)	Vision Template 3 (Localization)
Key	MS_SHA1	VS_SHA1	LS_SHA1	RS_SHA1	IS_SHA1	CD_SHA1	VL1_SHA1	VL2_SHA1	VL3_SHA1

	Vision Template 4 (Localization)	Vision Template 5 (Recheck)	Vision Template 6 (Recheck)	Vision Template 7 (Recheck)	Vision Template 8 (Recheck)	Vision Template 9 (Recheck)	Vision Template 10 (Recheck)	Vision Template 11 (Recheck)	Vision Template 12 (Recheck)
Key	VL4_SHA1	VR1_SHA1	VR2_SHA1	VR3_SHA1	VR4_SHA1	VR5_SHA1	VR6_SHA1	VR7_SHA1	VR8_SHA1

Audio/Home SW DFM v2.8

Machine Data

```
{
  “unit_sn ”: “xxxxx”,                Standard SN. No personalized suffix or prefix.
  “serials ”: {                       Put all non-parametric data in “serials”(For example, module SN, conveyor location, glue SN, etc. Do not include “sw_version” and “limits_version”)
    “mac_addr ”: “C4:00:AD:7F:A8:36”, Get and upload the Mac address for HIVE network
    “top_case ”: “xxxxx”,
    “bottom_case ”: “xxxxx”,          Use when a machine has more than 1 conveyor. Especially for dispenser. Facing the input side, “L” for left conveyor. “R” for right conveyor.
    “conveyor”: “L”,                  Only for dispenser. To record and upload current glue SN and purge id.
    “glue_sn”: “AAAA1234”,
    “purge_idx”: 20
    }
  “pass”: “true”,                     Whether the unit is passed or failed, machine data with SN should always be uploaded to HIVE as a record. Value: “true” or “false”.
  “input_time”: “2022-02-01T12:59:05.06+0800”,
  “output_time”: “2022-02-01T12:59:10.23+0800”,
  “data”: {                           Only put parametric data “sw_version” and “limits_version” here. Parametric means its value is numeric
    “hive_state”: 1,                  Auto get current HIVE state and upload along with each SN
    “output_ct”: 10,
    “unit_ct”: 10,
    “sw_version”: “xxxxx”,            HIVE reserved keys. Must be exactly the same
    “limits_version”: “xxxxx”
    }
}
```

Audio/Home SW DFM v2.8

Error Data

```
{
    "message": "xxxxx",
    "code": "123",
    "severity": "critical",
    "occurrence_time ": "2022-02-01T12:59:05.06+0800",
    "resolved_time ": "2022-02-01T12:59:10.23+0800",
    "data": {
        "hive_state": 1,
        "error_detail": "xxxxx"
        }
}
```

Follow error type definition

Choose from "warning" / "error" / "critical"

Follow error type definition

Audio/Home
SW DFM v2.8

S2.3-HIVE/Insight Integration

Send pdata and blob data to Insight(Machine data only)

ONLY FOR STATIONS SENDING DATA TO INSIGHT: PARAMETRIC/ATTRIBUTE/LIMIT DATA AND BLOB FILE

Audio/Home
SW DFM v2.8

Machine Data

When enabling HIVE to upload pdata and blob to Insight, use ***multipart/form-data*** as ***Content-Type***

!!!!!!!! NOT ***application/json***

```
{
    "unit_sn ": "xxxxx",
    "serials ": {
        "top_case ": "xxxxx",
        "bottom_case ": "xxxxx",
        "conveyor": "L",
        "glue_sn": "AAAA1234"
    }
    "pass": "true",
    "input_time": "2022-02-01T12:59:05.06+0800",
    "output_time": "2022-02-01T12:59:10.23+0800",
    "data": {
        "fixture_id": "test_fixture",
        "head_id": "test_head",
        "hive_state": 1,
        "output_ct": 10,
        "sw_version": "xxxxx",
        "limits_version": "xxxxx",
        "measurementA ": 0.56,
        "measurementB": 0.55,
        "measurementC": 0.51,
        "measurementD": 0.65
    }
    "limit": {
        "measurementA": {
            "upper_limit": 1.0,
            "lower_limit": 0.1
        },
        "measurementB": {
            "upper_limit": 1.0
        }
        "measurementC": {
            "lower_limit": 0.1
        }
    }
    "blob": {
        "file_name": "image.jgp"
    }
}
```

For all non-numerical data required in traceability matrix, add them here

HIVE reserved keys. Must be exactly the same

Replace test_fixture with values of Fixture ID value

Replace test_head with values of Test Head

For all numerical data required in traceability matrix, add them here

For all numerical data with have upper or lower limit, add them here. All keys in limit must also appears in "data"

Fixture ID	Test Head
KSH36XNP1LH900XXSUX100003	1

Example message sent from Mac or Linux command line **Without blob data**

```
curl -X POST -H "Content-Type: multipart/form-data"
http://10.0.0.2:5008/v5/capture/machinedata -F 'jsondata={"unit_sn":"SN000000001","serials":
{"top_case":"AB05684AZ","bottom_case":"67BA099"},"pass":"true", "input_time":"2019-05-
24T09:22:59.784515+0000","output_time":"2019-05-24T09:23:59.792048+0000","data":{"CDMA":
"0.00"},"limit": {"MeasurementA":{      "upper_limit":1.0,"lower_limit":0.1},"MeasurementB":
{"upper_limit":1},"MeasurementC":{"lower_limit":0}},"attr":
{"Attr1":"Green","Attr2":"Factory"},"blobs":[]}'
```

Keep this even when no blob data

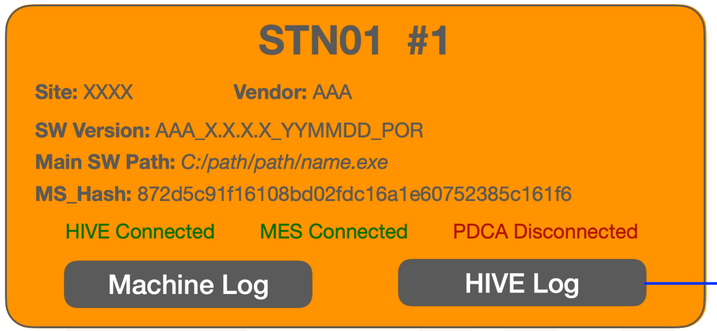
Example message sent from Mac or Linux command line **With blob data**

```
curl -X POST -H "Content-Type: multipart/form-data"
http://10.0.0.2:5008/v5/capture/machinedata -F 'jsondata={"unit_sn":"SN000000001","serials":
{"top_case":"AB05684AZ","bottom_case":"67BA099"},"pass":"true", "input_time":"2018-01-
16T13:59:05.06+0800","output_time":"2018-01-16T13:59:25.06+0800","data":
{"MeasurementA":0.56,"MeasurementB":0.55,"MeasurementC":0.5466},"limit": {"MeasurementA":{
"upper_limit":1.0,"lower_limit":0.1},"MeasurementB":{"upper_limit":1},"MeasurementC":
{"lower_limit":0}},"attr":{"Attr1":"Green","Attr2":"Factory"},"blobs":
[{"file_name":"test.log"},{"file_name":"image.jpg","retention_policy":180}]}' -F
'file1=@/home/test/test.log' -F 'file2=@/home/test/image.jpg'
```

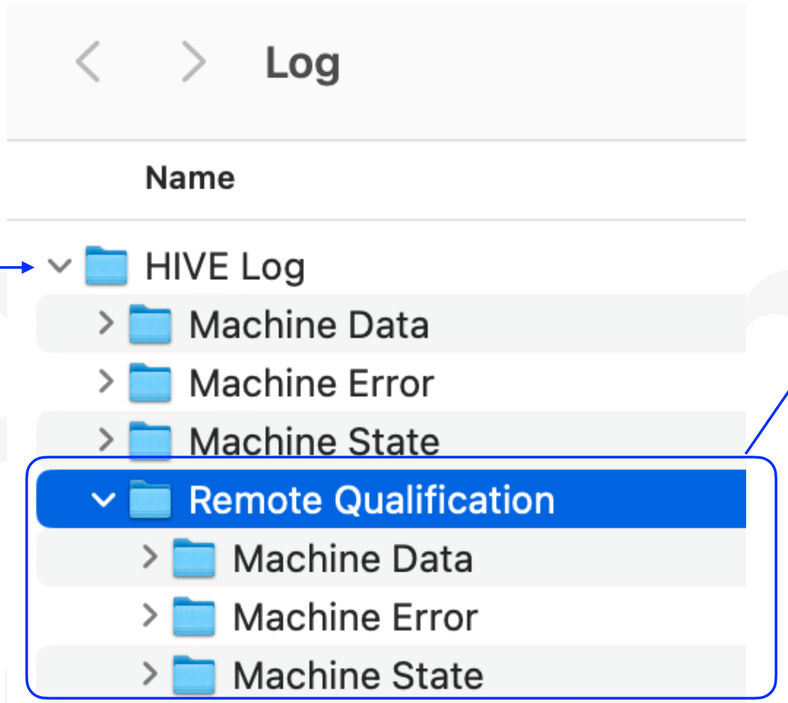
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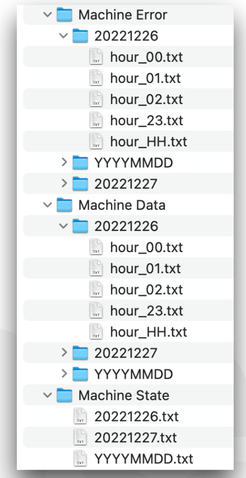
S2.4-Extra Log for Hive Remote Qualification (When Remote Qualification is enabled)

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- HIVE log button will link to HIVE log, which contains Remote Qualification log





Key Requirements

- Separate folder under HIVE log
- Plain json data only
- 1 full json script per line

Example

MachineState.txt

```
{"machine_state":1,"state_change_time": "2023-02-01T01:31:03.07+0800","data": {"VL1_SHA1":"bda4cb9bebfbc78a4681280737cf998d28990e2a"}}
{"machine_state":2,"state_change_time": "2023-02-01T02:46:08.86+0800","data": {"VL1_SHA1":"bda4cb9bebfbc78a4681280737cf998d28990e2a"}}
{"machine_state":1,"state_change_time": "2023-02-01T02:55:55.27+0800","data": {"VL1_SHA1":"bda4cb9bebfbc78a4681280737cf998d28990e2a"}}
{"machine_state":2,"state_change_time": "2023-02-01T02:57:41.52+0800","data": {"VL1_SHA1":"bda4cb9bebfbc78a4681280737cf998d28990e2a"}}
```

Full json script per line. No line break in between

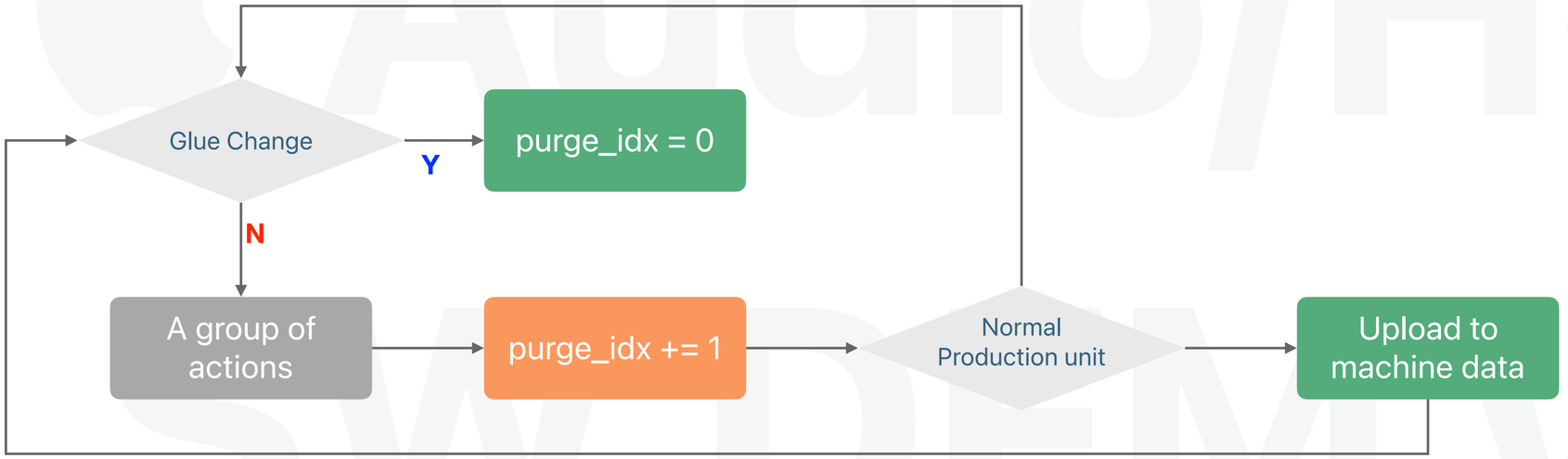
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S2.5-Dispenser purge_index collection with HIVE

Audio/Home
SW DFM v2.8

Purpose: To track the purge frequency of each volume of glue

- Reset to 0 after glue change
- +1 after each group of dispensing or purge
  - A group of actions = Dispense all glue path for a unit.
  - A group of actions = Every purge of glue during valve idle time
  - A group of actions = Every purge of glue when do glue weighing




```
{
    “unit_sn ”: “xxxxx”,
    “serials ”: {
        “top_case ”: “xxxxx”,
        “bottom_case ”: “xxxxx”,
        “conveyor”: “L”,
        “glue_sn”: “AAAA1234”,
        “purge_idx_front”: 20,
        “purge_idx_rear”: 20
        }
    “pass”: “true”,
    “input_time”: “2022-02-01T12:59:05.06+0800”,
    “output_time”: “2022-02-01T12:59:10.23+0800”,
    “data”: {
        “hive_state”: 1,
        “output_ct”: 10,
        “sw_version”: “xxxxx”,
        “limits_version”: “xxxxx”
        }
}
```


unit_sn	Conveyor	purge_idx
SN1	L	2
SN2	L	3
SN3	R	3
SN4	L	10
SN5	R	4
SN6	R	5

- SN1, SN2, SN4 went through left conveyor
- No extra purge between dispensing on SN1 and SN2
- 7 extra purge between dispensing SN2 and SN4

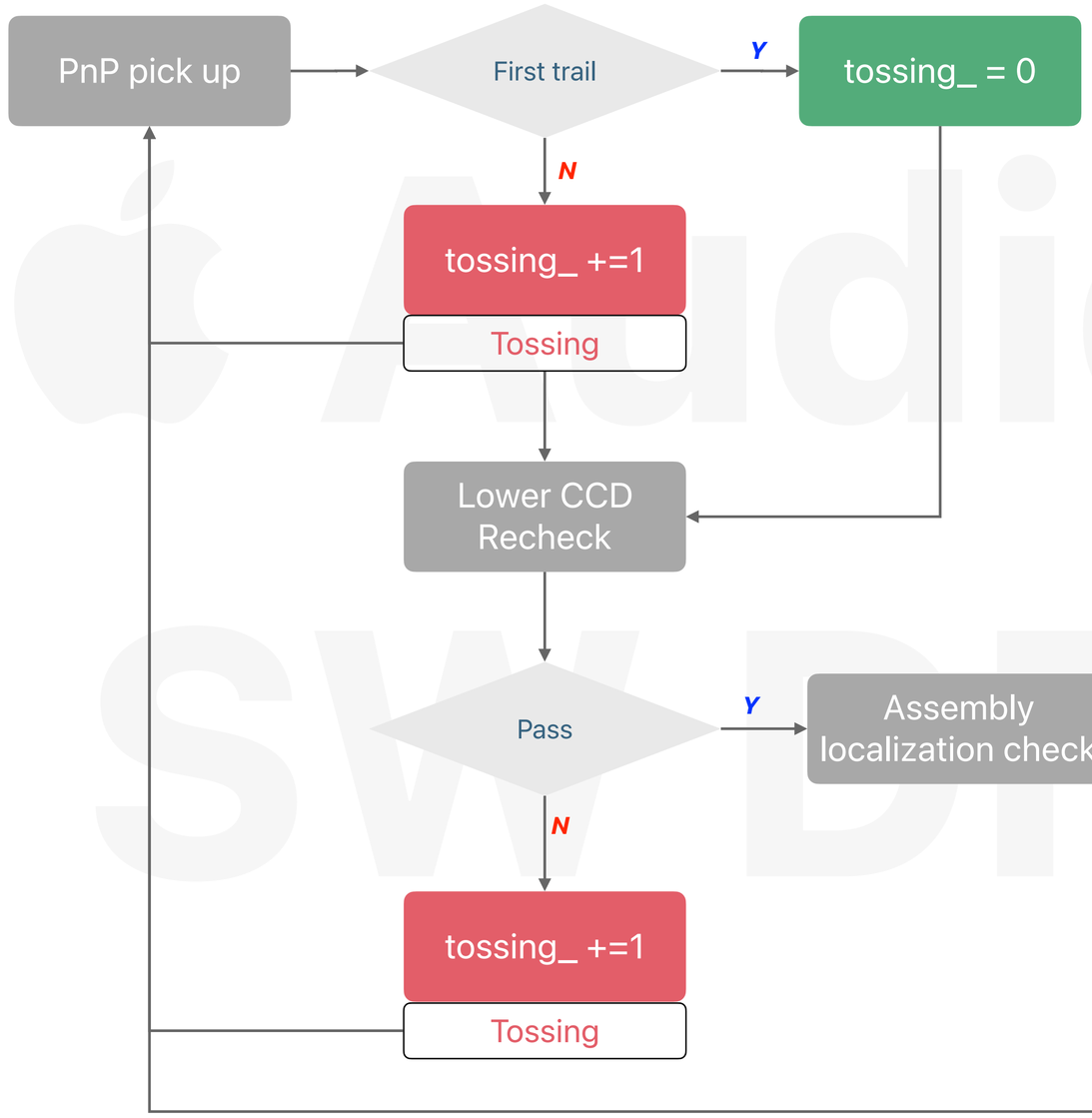
- SN3, SN5, SN6 went through right conveyor
- No extra purge between dispensing on SN3, SN5 and SN6

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S2.6-Tossing Data Collection

Audio/Home
SW DFM v2.8

Tossing data collection for **PnP**




```
{
  "unit_sn ": "1234567890",
  "serials ": {
    "sub_id": "S1",        Upload only when machine has sub-stations. Use "S1;S2;S3" is more than one sub-station
    "error_message": "Tossing Error",
    "error_detail ": "tossing detail",        Upload "Message1;Message2;Message3" if more than one error description
    },
  "pass": "true/false",        Final assemble result
  "input_time": "2022-02-01T12:59:05.06+0800",
  "output_time": "2022-02-01T12:59:10.23+0800",
  "data": {
    "hive_state": 1,
    "output_ct": 10,
    "tossing_1": 3,        Add tossing_2, tossing_3, etc. if more than 1 material to be pick and placed to units
    "sw_version": "xxxxx",
    "limits_version": "xxxxx"
    }
}
```

unit_sn	_pass	tossing_1
Actual SN1	TRUE	0
Actual SN2	FALSE	1
Actual SN3	TRUE	2
Actual SN4	FALSE	4
Actual SN5	FALSE	0
Actual SN5	TRUE	1

Tossing rate = 61.54%=(0+1+2+4+0+1)/(**5**+0+1+2+4+0+1): 8x dropped, 5x assembled

Retest rate = 20%=1/5

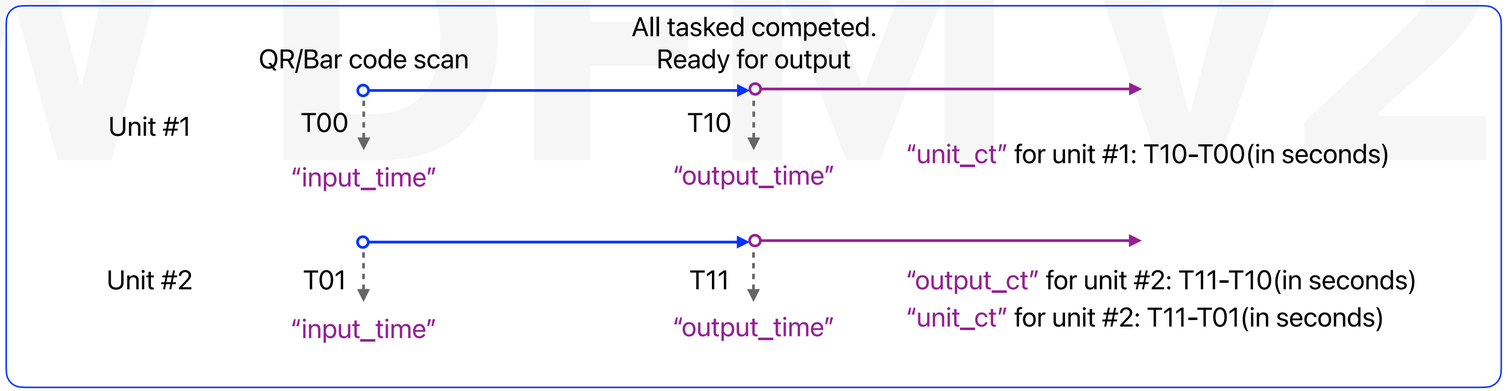
Apple Confidential

S2.7-CT Collection with HIVE

Audio/Home
SW DFM v2.8

Machine Data

```
{
  “unit_sn ”: “xxxxx”,                    Standard SN. No personalized suffix or prefix.
  “serials ”: {                           Put all non-parametric data in “serials”(For example, module SN, conveyor location, glue SN, etc. Do not include “sw_version” and “limits_version”)
    “top_case ”: “xxxxx”,
    “bottom_case ”: “xxxxx”,
    “conveyor”: “L”,                      Use when a machine has more than 1 conveyor. Especially for dispenser. Facing the input side, “L” for left conveyor. “R” for right conveyor.
    “glue_sn”: “AAAA1234”,                Only for dispenser. To record and upload current glue SN and purge id.
    “purge_idx”: 20
    }
  “pass”: “true”,                         Whether the unit is passed or failed, machine data with SN should always be uploaded to HIVE as a record. Value: “true” or “false”.
  “input_time”: “2022-02-01T12:59:05.06+0800”,
  “output_time”: “2022-02-01T12:59:10.23+0800”,
  “data”: {                               Only put parametric data “sw_version” and “limits_version” here. Parametric means its value is numeric
    “hive_state”: 1,                      Auto get current HIVE state and upload along with each SN
    “output_ct”: 10,
    “unit_ct”: 10,                        HIVE reserved keys. Must be exactly the same
    “sw_version”: “xxxxx”,
    “limits_version”: “xxxxx”
    }
}
```



*Any failure during processing a unit will trigger machine to send data with SN to HIVE with key “pass” valued as “false”. Using failure timestamp as “output_time”

Exceptions:

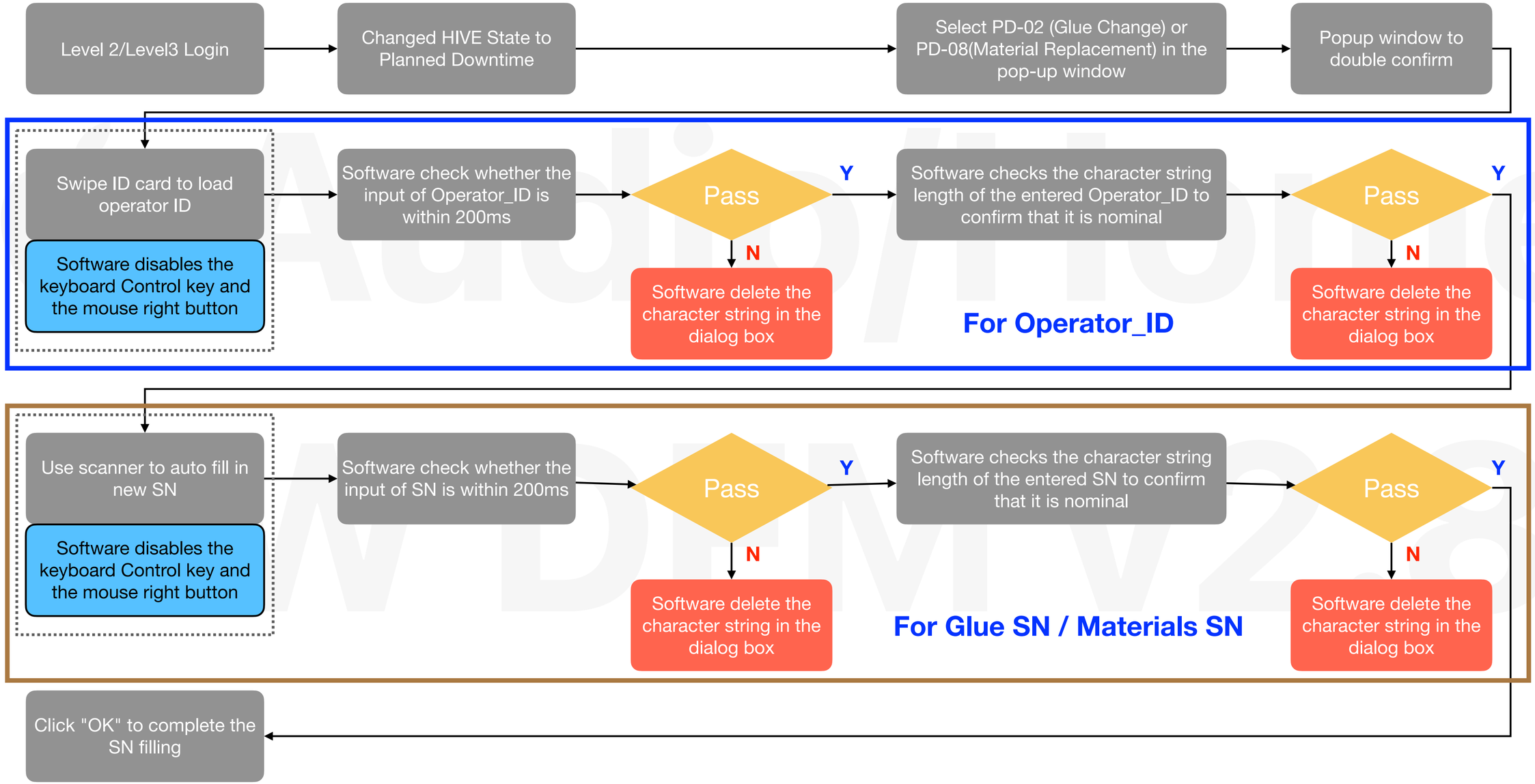
- Upload 0 as CT for first unit
- Maximum value is 3600

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S2.8-SW Logic to avoid manually key-in ID and SN

Audio/Home
SW DFM v2.8

Fill in SN(glue/materials) to Software for **PnP & SCUD**

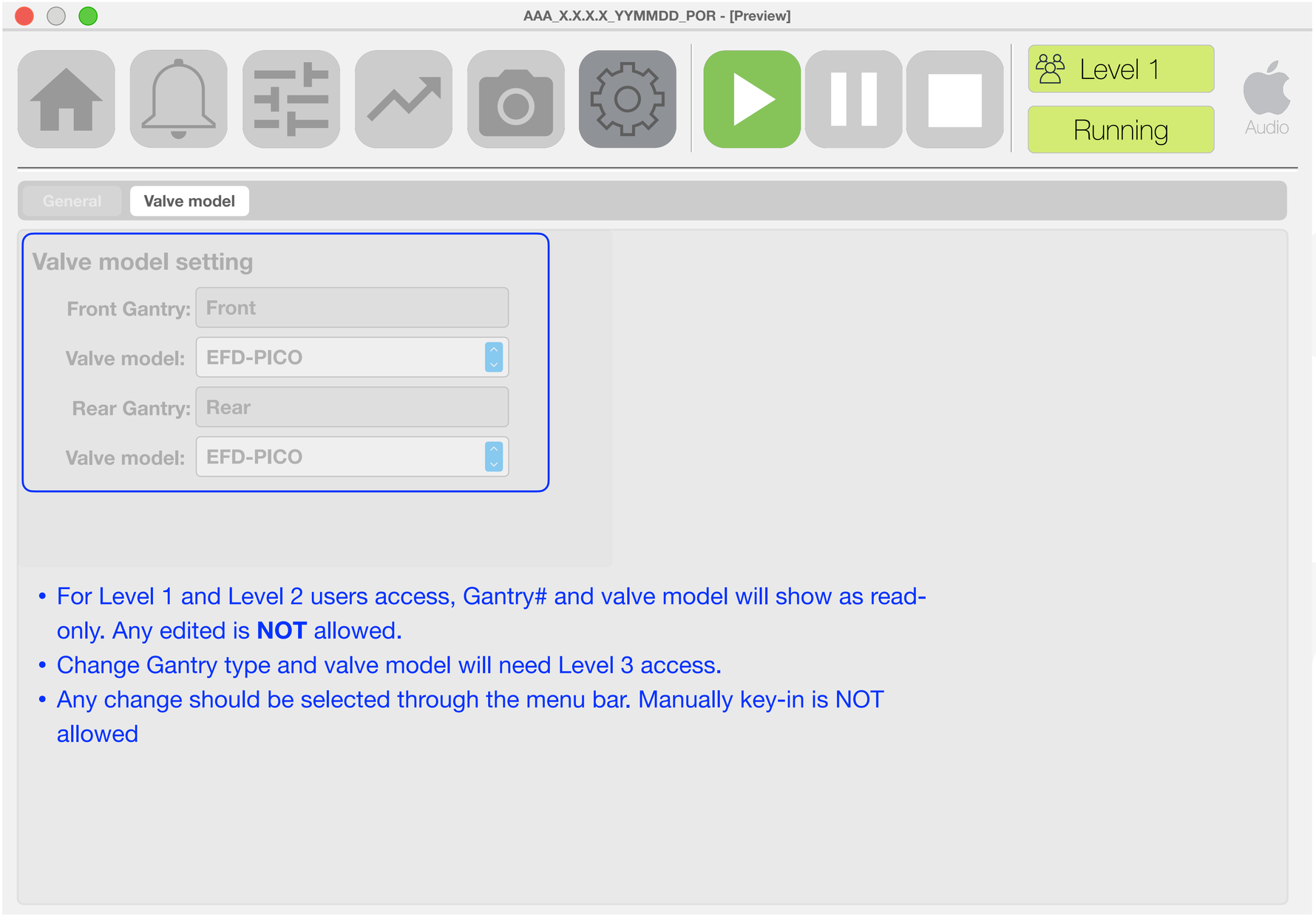


S3-Planned Downtime Process Flow

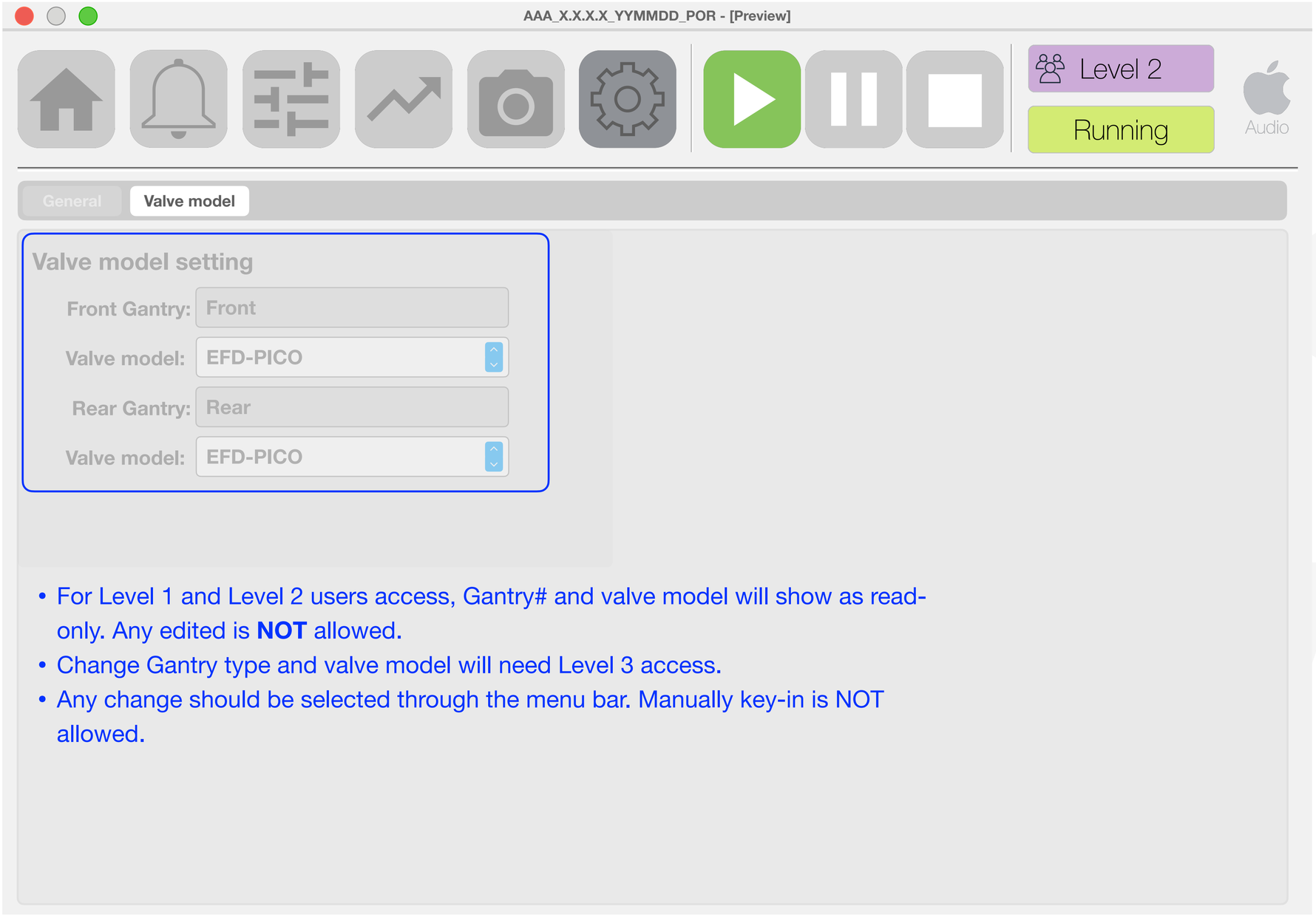
Audio/Home
SW DFM v2.8

S3.1-Valve Change (PD-04)

Audio/Home
SW DFM v2.8

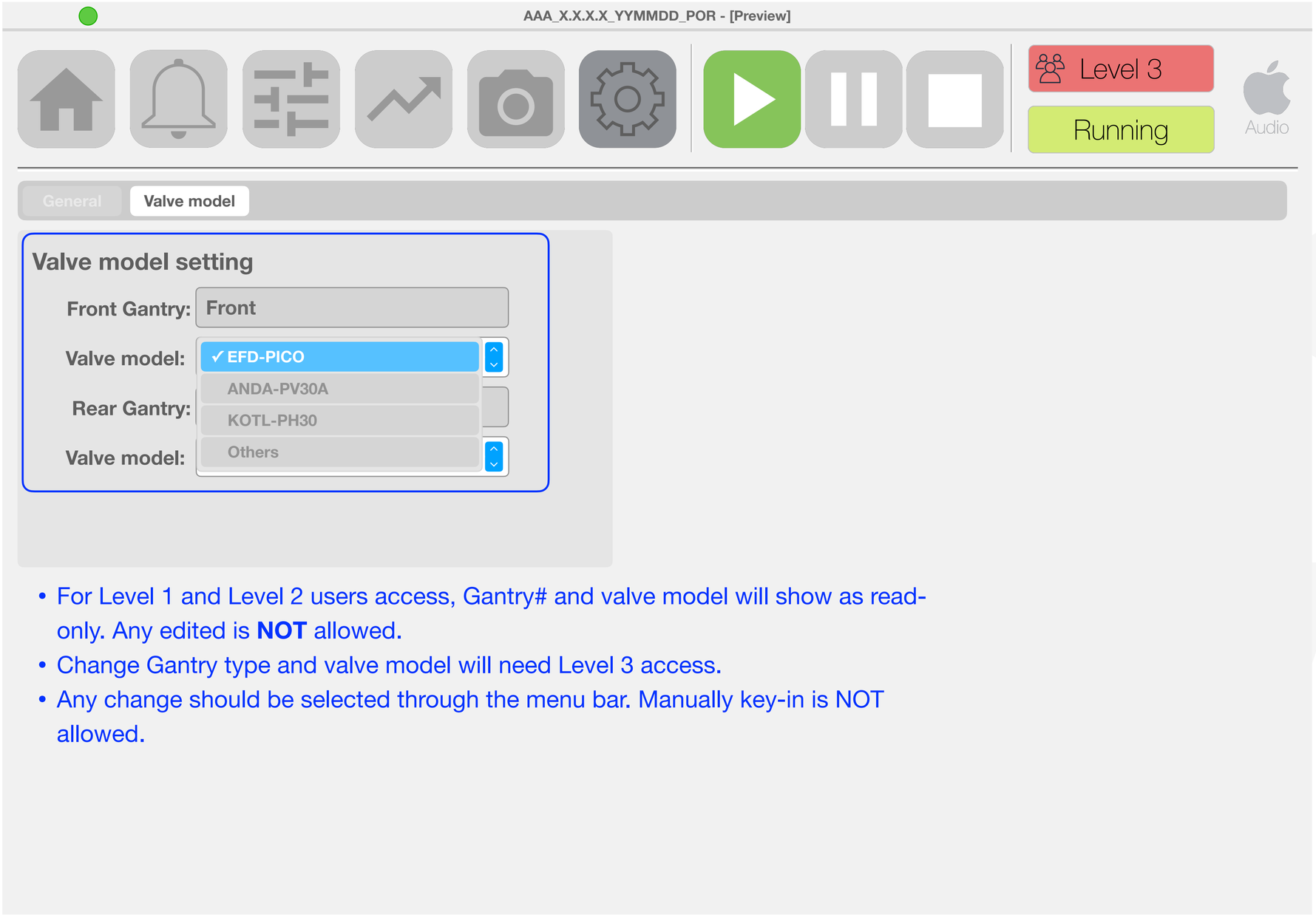


- For Level 1 and Level 2 users access, Gantry# and valve model will show as read-only. Any edited is **NOT** allowed.
- Change Gantry type and valve model will need Level 3 access.
- Any change should be selected through the menu bar. Manually key-in is NOT allowed



- For Level 1 and Level 2 users access, Gantry# and valve model will show as read-only. Any edited is **NOT** allowed.
- Change Gantry type and valve model will need Level 3 access.
- Any change should be selected through the menu bar. Manually key-in is NOT allowed.

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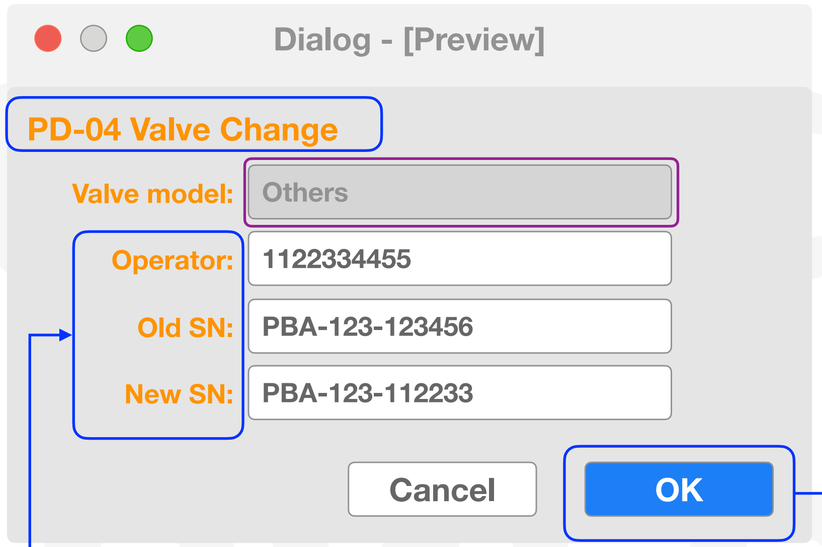
- For Level 1 and Level 2 users access, Gantry# and valve model will show as read-only. Any edited is **NOT** allowed.
- Change Gantry type and valve model will need Level 3 access.
- Any change should be selected through the menu bar. Manually key-in is NOT allowed.

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★For PD-04 Valve Change

When entering ANY state by manually, popup window for filling operator/scan the valve SN info and record the SN to the configuration file, and upload the valve SN info to HIVE/MES

For planned downtime that are related with a physical SN



- Swipe ID card to load operator ID
- Old SN will show as recorded
- Use scanner to auto fill in new SN. Manually key in is **NOT** allowed
  - ❖ Scanning and filling valve SN, the software automatic check valve SN's string contains the following content. If it contains them, the software automatically deletes the character in the dialog box and prompts "Valve model selection error, pls select the correct valve model on the settings page".
    - ✤ EFD-PICO
    - ✤ ANDA-PV30A
    - ✤ KOTL-PH30

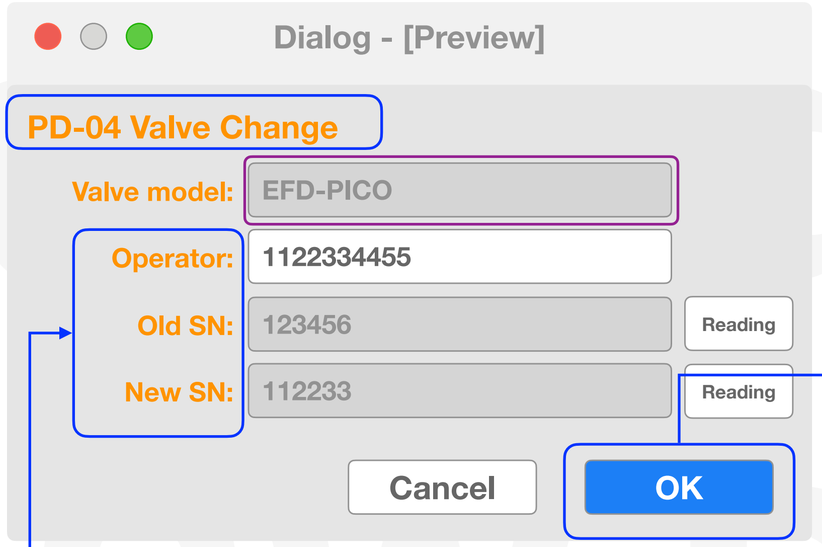
- Click the "OK" button and interact with the MES to get the valve usage information.
  - ❖ Successful: Update the new valve new SN and usage info data to configuration file and switch the HIVE state to Planned DT.
  - ❖ Error: Terminate the operation to change the valve (Do not update the valve new SN to configuration file. Do not change the HIVE status).
  - ❖ GET_RUBBERBODY_USED_INFO

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★For PD-04 Valve Change

When entering ANY state by manually, popup window for filling operator, readings valve SN info and record the SN to the configuration file, and upload the valve SN info to HIVE/MES

For planned downtime that are related with a Internal SN



- Swipe ID card to load operator ID
- Old SN will show as recorded
- The new SN will show as recorded and readings on the controller of the valve.

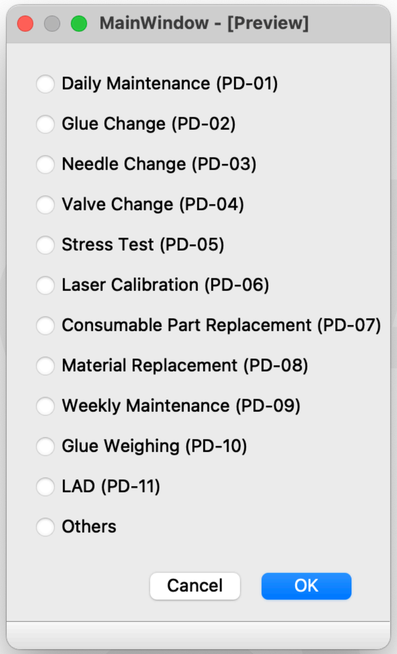
- After clicking on the "OK" button, obtaining the standard SN of the valve and the production tracking data interacts with the MES.
  - ❖ The API to get the valve standard SN: GET_RUBBERBODY_SN
    - ✤ If succeeds, update the standard SN of the new valve to the configuration file and execute another API to acquire the corresponding data.
    - ✤ If error, The next step cannot be executed, and an error message is displayed in the popup window.
  - ❖ The API to get valve usage info: GET_RUBBERBODY_USED_INFO
    - ✤ If succeeds: Update the new valve new SN and usage info data to configuration file and switch the HIVE state to Planned DT.
    - ✤ If error: Terminate the operation to change the valve (Do not update the valve new SN to configuration file. Do not change the HIVE status).

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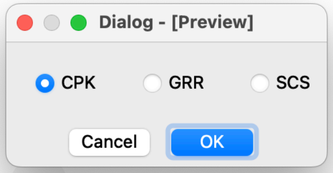
S3.2-LAD with HIVE(PD-11)

Audio/Home
SW DFM v2.8

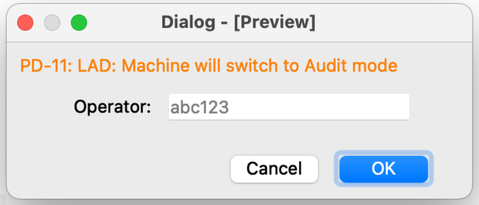
LAD Testing Mode



Choose planned downtime -> LAD before starting LAD test



Popup window to confirm



Machine switch to planned DT after scanning badge card and confirming



- For LAD testing, upload extra keys in defend format
- Keys and values are LAD only

LAD Keys	Mode	Operator_ID	TestSeriesID
CPK	1	1	yyyyMMddHHmmss
GRR	2	1 or 2 or 3	yyyyMMddHHmmss
SCS	3	1	yyyyMMddHHmmss

Machine Data Example

Keys are caps sensitive

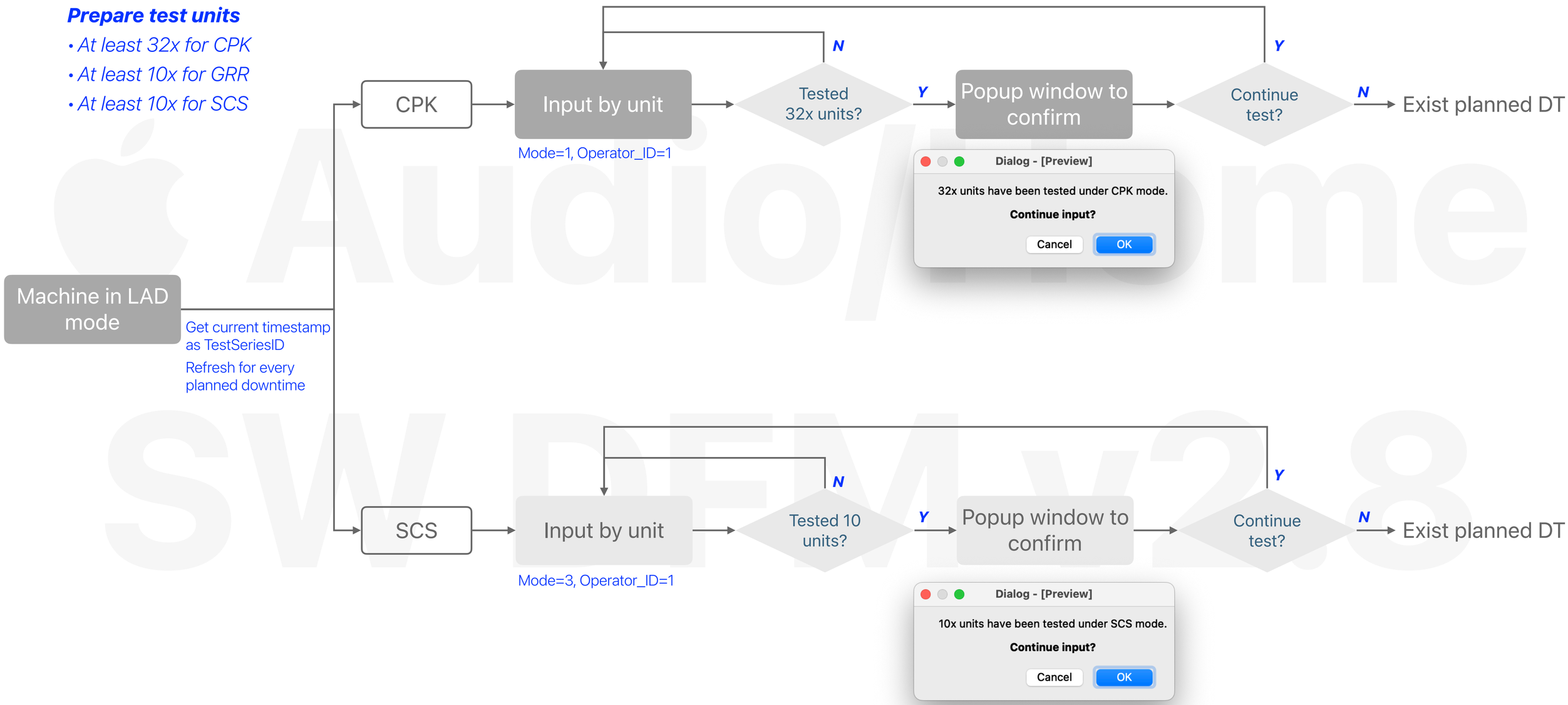
```
{
    “unit_sn ”: “xxxxx”,
    … …
    “data”: {
        … …
        “audit”: true,
        “Mode”: 1,
        “Operator_ID”: 1,
        “TestSeriesID”:  20230523195023,
        … …
        }
        … …
}
```

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LAD Testing Flow

Prepare test units

- *At least 32x for CPK*
- *At least 10x for GRR*
- *At least 10x for SCS*

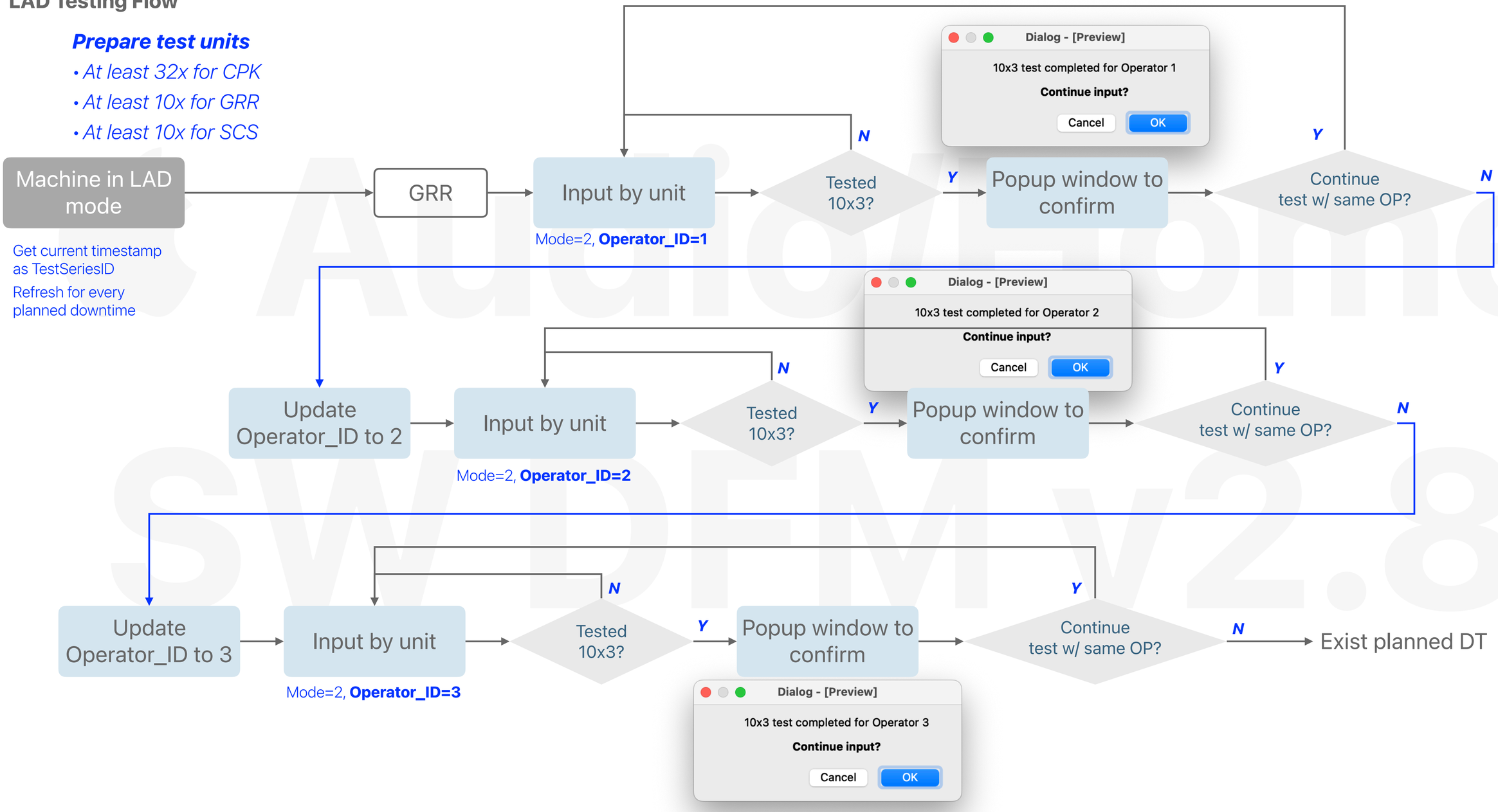


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LAD Testing Flow

Prepare test units

- *At least 32x for CPK*
- *At least 10x for GRR*
- *At least 10x for SCS*



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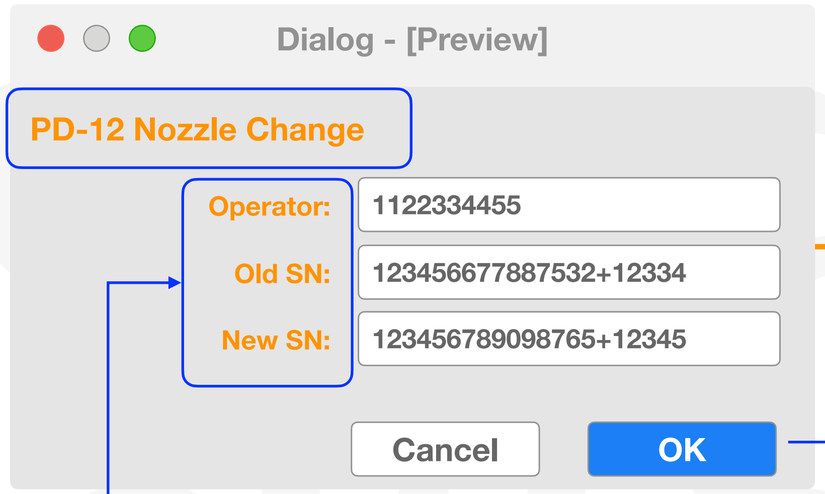
S3.3- Nozzle Change (PD-12)

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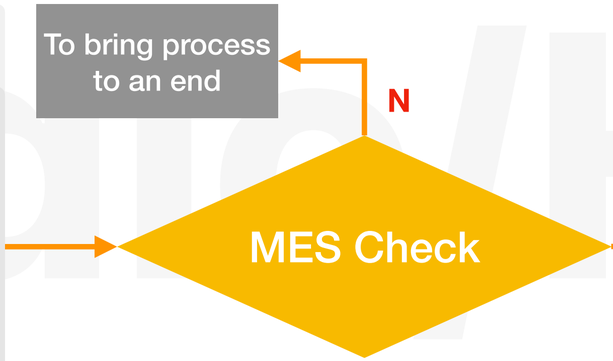
★For Nozzle Change

Popup window for filling operator id/Nozzle SN info.

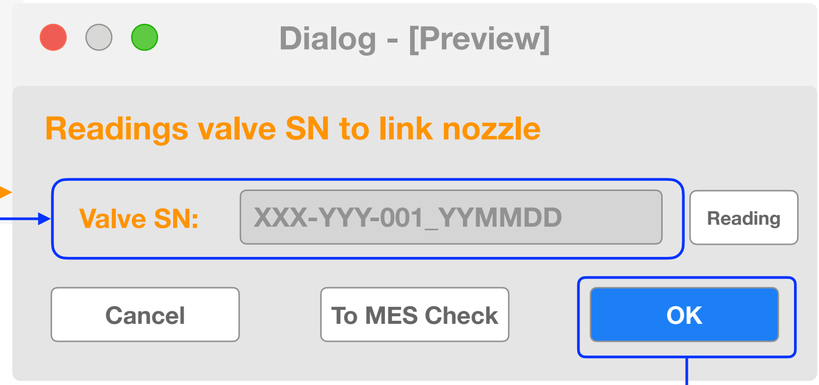
For planned downtime that are related with a physical SN



- Swipe ID card to load UID
- Old SN will show as recorded
- Use keyboard to manually key-in new SN.



Popup window to link nozzle (next page)



- The Valve standard SN will show as recorded and readings on the configuration file.

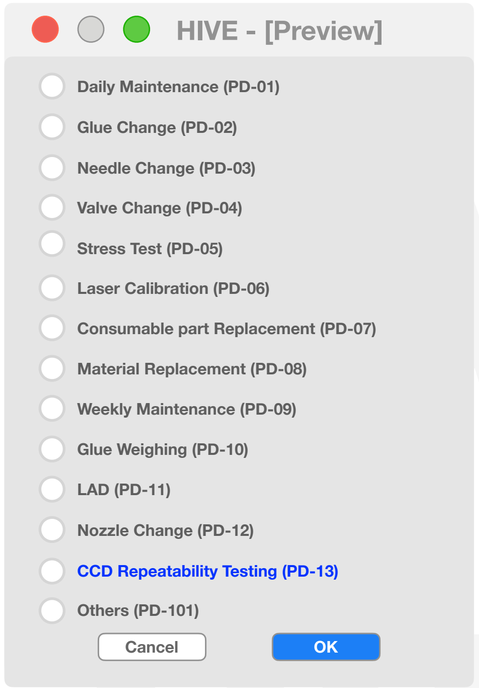
- The Valve standard SN will show as recorded and readings on the configuration file.
  - ❖ Read the configuration file to get the valve standard SN. After clicking the "To MES Check" button, the machine send a link request to the MES for check and link.
    - ✤ MES check “OK” and return “true” message, enable the "OK" button.
    - ✤ MES check “NG” and return “false” message, disable the "OK" button.
  - ❖ If read the valve SN failure. End the process and prompt the corresponding message.

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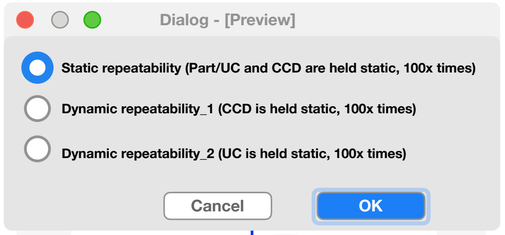
S3.4-CCD Repeatability Testing(PD-13)

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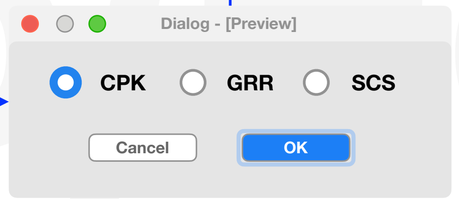
CCD Repeatability Testing Mode



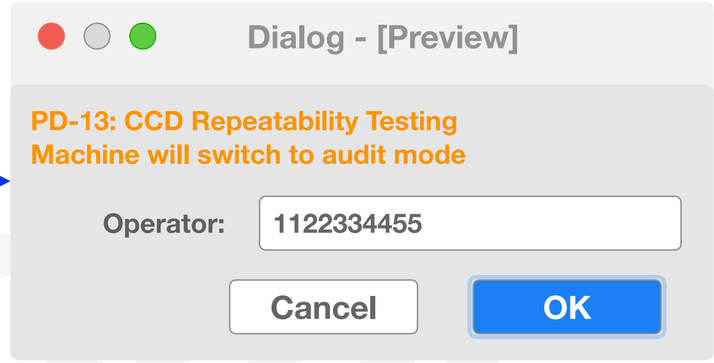
Choose planned downtime -> CCD Repeatability Testing before starting test



Popup window to confirm



Popup window to confirm



Machine switch to planned DT after scanning badge card and confirming



- For LAD testing, upload extra keys in defend format
- Keys and values are LAD only

LAD Keys	Mode	Operator_ID	TestSeriesID
CPK	1	1	yyyymmddhhmmss
GRR	2	1 or 2 or 3	yyyymmddhhmmss
SCS	3	1	yyyymmddhhmmss

Machine Data Example

Keys are caps sensitive

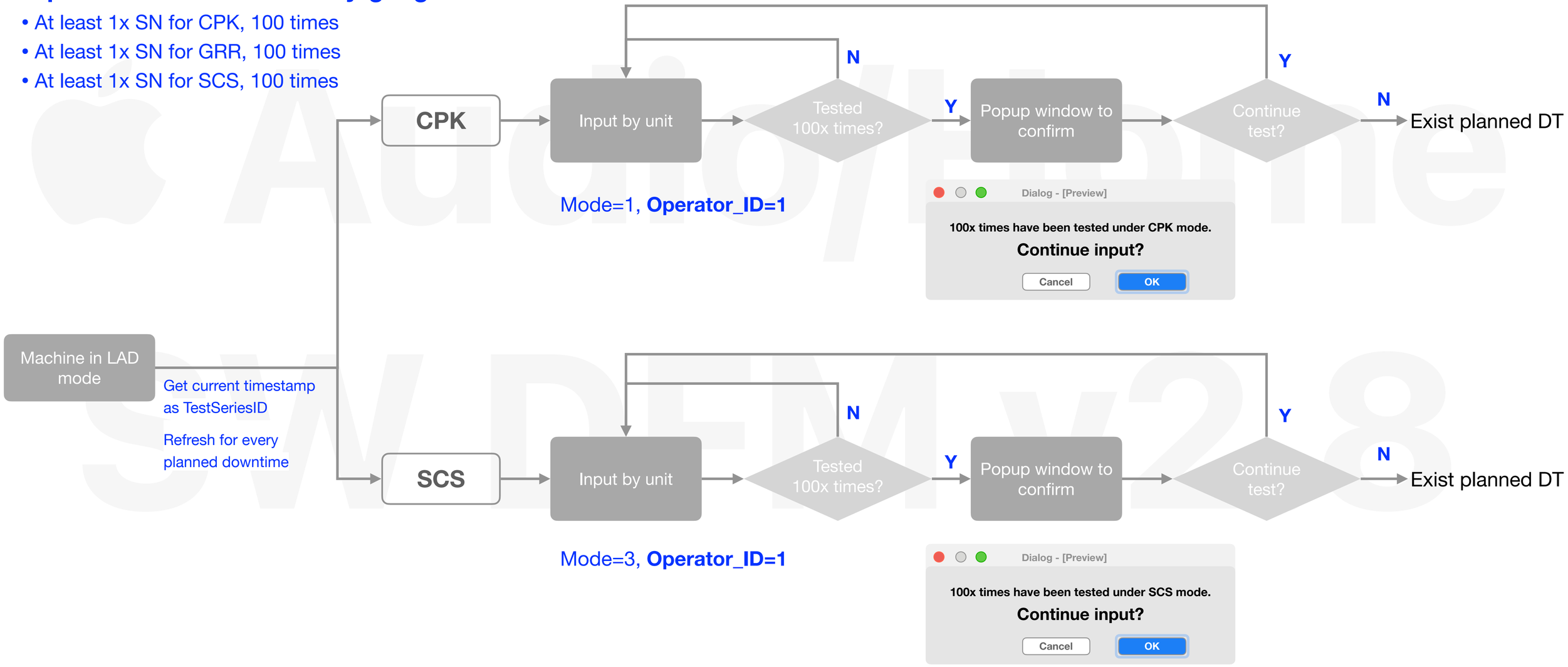
```
{
    “unit_sn”:”xxxxx”,
        … …
    “data”:{
        … …
        “audit”: true,
        “Mode”: 1,
        “Operator_ID”: 1,
        “TestSeriesID”: 20230523195023,
        … …
        }
}
```

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CCD Repeatability Testing Flow

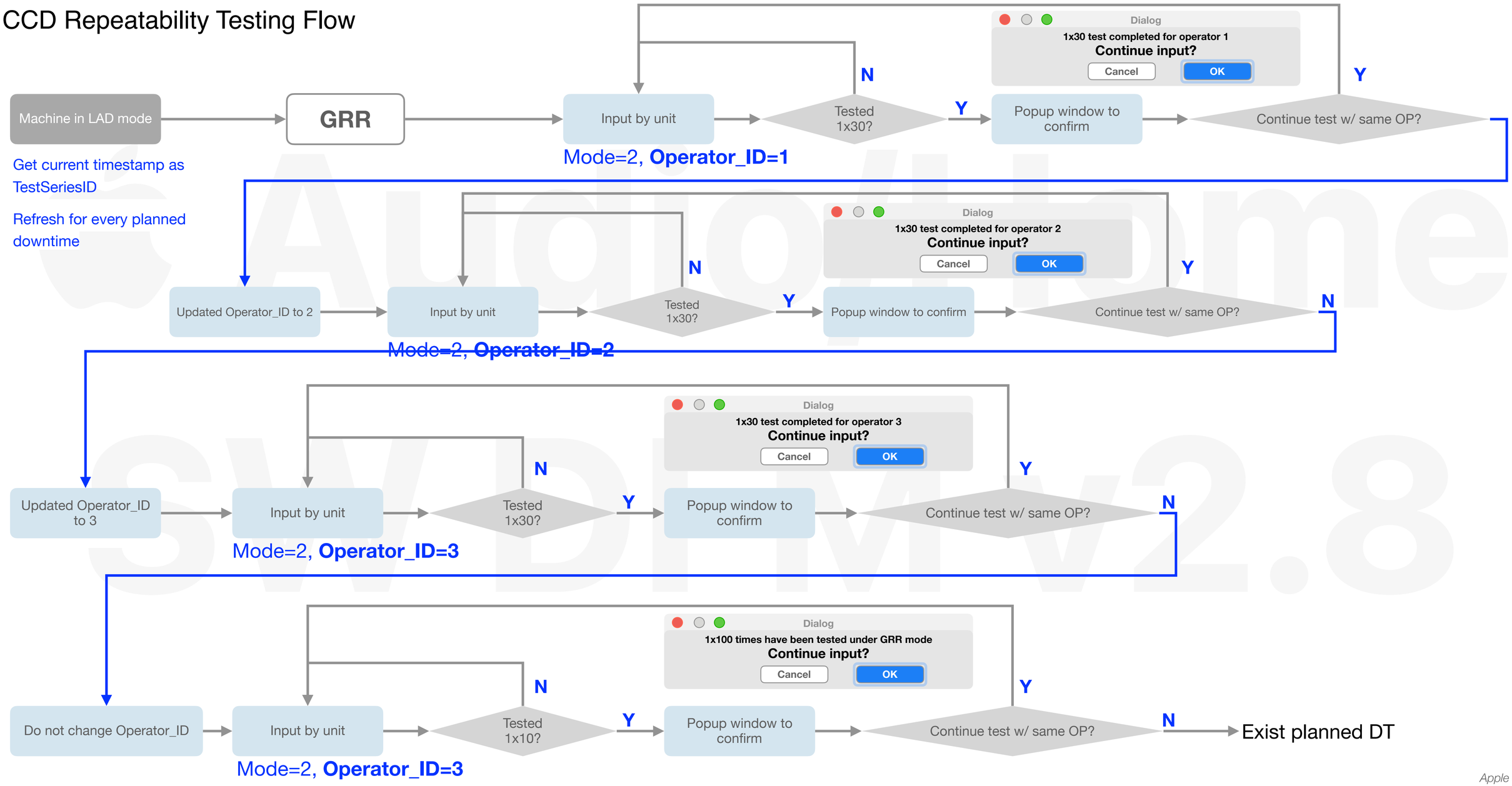
Prepare test UC&unit/dummy/gauge block

- At least 1x SN for CPK, 100 times
- At least 1x SN for GRR, 100 times
- At least 1x SN for SCS, 100 times



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CCD Repeatability Testing Flow

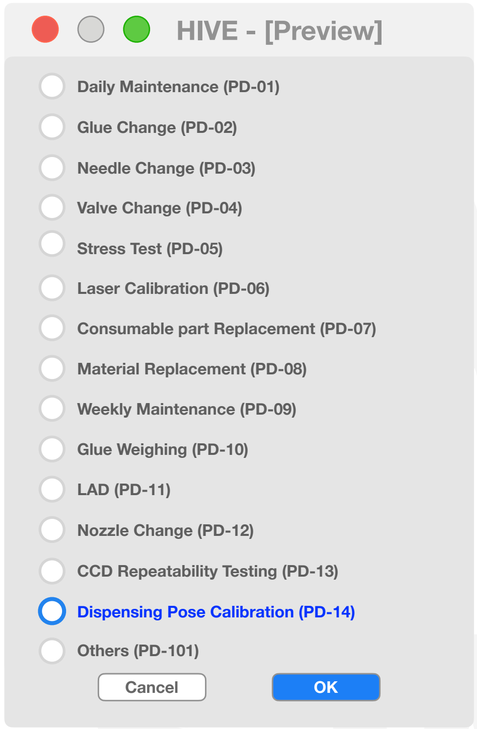


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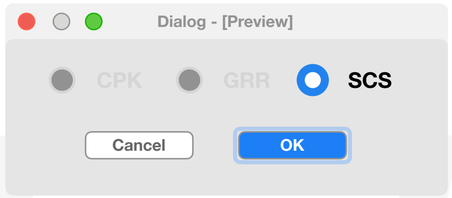
S3.5-Dispensing Pose Calibration(PD-14)

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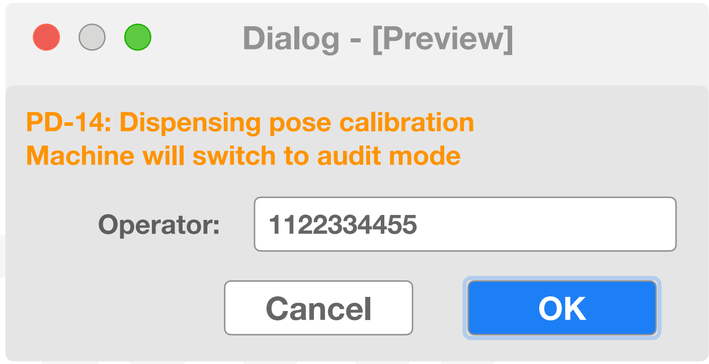
Dispensing Pose Calibration Mode



Choose planned downtime -> Dispensing pose calibration



Popup window to confirm



Machine switch to planned DT after scanning Card UID and confirming (Swipe badge card to auto read UID. Manually key in is **NOT** allowed)

!!! Only enable when Admin user logged in



- For LAD testing, upload extra keys in defend format
- Keys and values are LAD only

LAD Keys	Mode	Operator_ID	TestSeriesID
CPK	1	1	yyyymmddhhmmss
GRR	2	1 or 2 or 3	yyyymmddhhmmss
SCS	3	1	yyyymmddhhmmss

Machine Data Example

Keys are caps sensitive

```
{
    “unit_sn”:”xxxxx”,
        … …
    “data”:{
        … …
        “audit”: true,
        “Mode”: 3,
        “Operator_ID”: 1,
        “TestSeriesID”: 20230523195023,
        … …
        }
}
```

For Dispensing Pose Calibration

The requirements for dispensing pose calibration are as follows:

- Flatness and dispensing pose calibration must be fully automated.
- The settings related to flatness and dispensing pose are critical parameters, and L3 user access is required to change them, including but not limited to the following settings and actions:
  - UI operations.
  - Axis speed settings.
  - Axis position settings.
  - Flatness nominal and SPEC limits settings.
  - Dispensing post nominal and setting of SPEC limits.
- For machines with dual gantry and dual conveyor, upload data to HIVE/Insight after performing all calibration steps.
- During the execution of the calibration, any calibration value out of specification terminates the calibration process. No data is uploaded to HIVE, and a pop-up window is displayed to remind OSS to check and fine-tune the machine.
- Any manual change of HIVE status during this period should be disabled. If the machine is down during the calibration period, the calibration process needs to be restarted.
- Upload parameter names are named according to established rules.
- If the mounting method of the laser module needs to be changed to enable calibration of flatness and dispensing pose, a pop-up window will prompt the OSS to change the mounting method after completing the calibration of one of the items before proceeding to the next step.
  - For example: Flatness calibration of the base is completed > pop-up window prompts > OSS clicks OK button > change the mounting method of the laser module > click on the software UI to calibrate the C-axis dispensing pose > pop-up window for secondary confirmation > calibrate C-axis pose...
- Compile a detailed operation SOP and train and test the OSS team to avoid any anomalies due to operational errors.

Audio/Home SW DFM v2.8



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